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(54) **ELECTRONIC DEVICE FOR DETECTING
SPAM CALL AND METHOD OF OPERATING
THE SAME**

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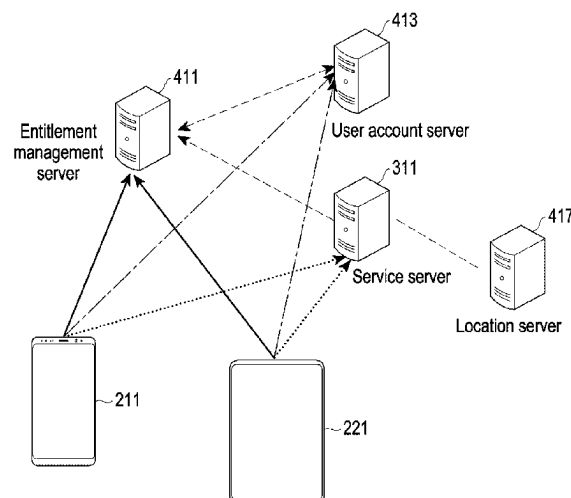
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(57) **ABSTRACT**

A server may include a communication circuit and at least one processor. The at least one processor may be connected to a first electronic device and a second electronic device, using the communication circuit, and may be configured to: obtain first call log information including at least one outgoing call made using a call function of the first electronic device by the second electronic device, determine whether the second electronic device satisfies a first condition for determining that the second electronic device is a spam caller, based on the first call log information, search for one or more external electronic devices located around the second electronic device, based on the first condition being satisfied, obtain second call log information about the one or more external electronic devices, and determine at least one of the one or more external electronic devices or the second electronic device as a spam caller, based on the second call log information.

16 Claims, 14 Drawing Sheets



(58) **Field of Classification Search**CPC ... H04M 2203/558; H04M 2203/6027; H04M
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See application file for complete search history.

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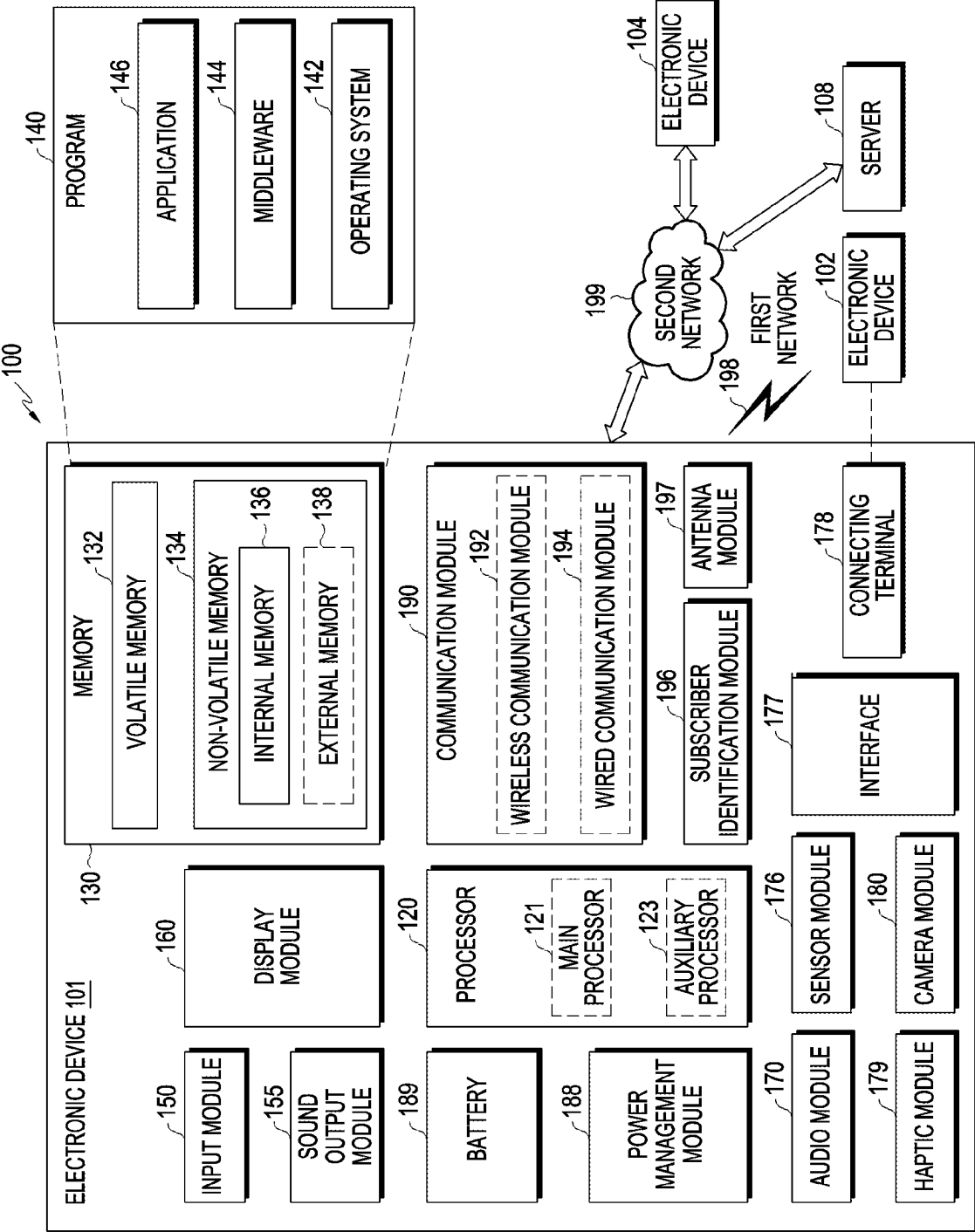


FIG. 1

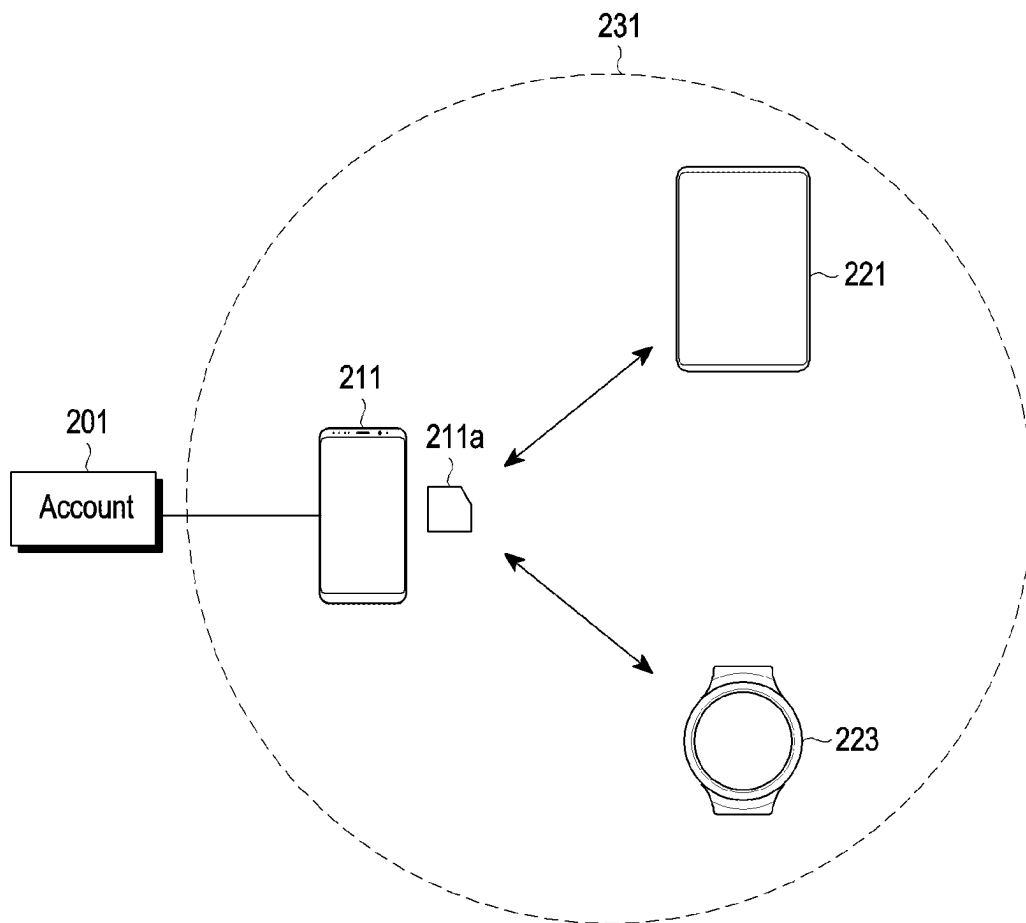


FIG. 2

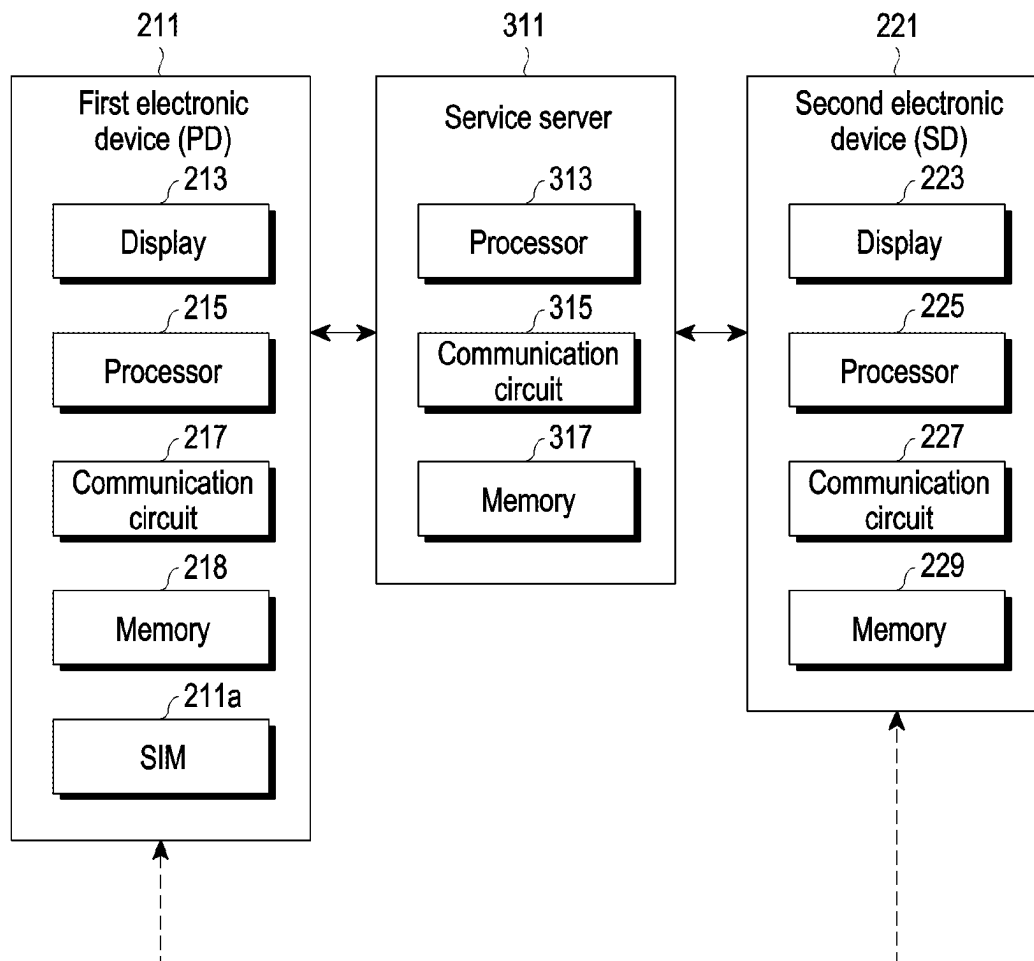


FIG. 3

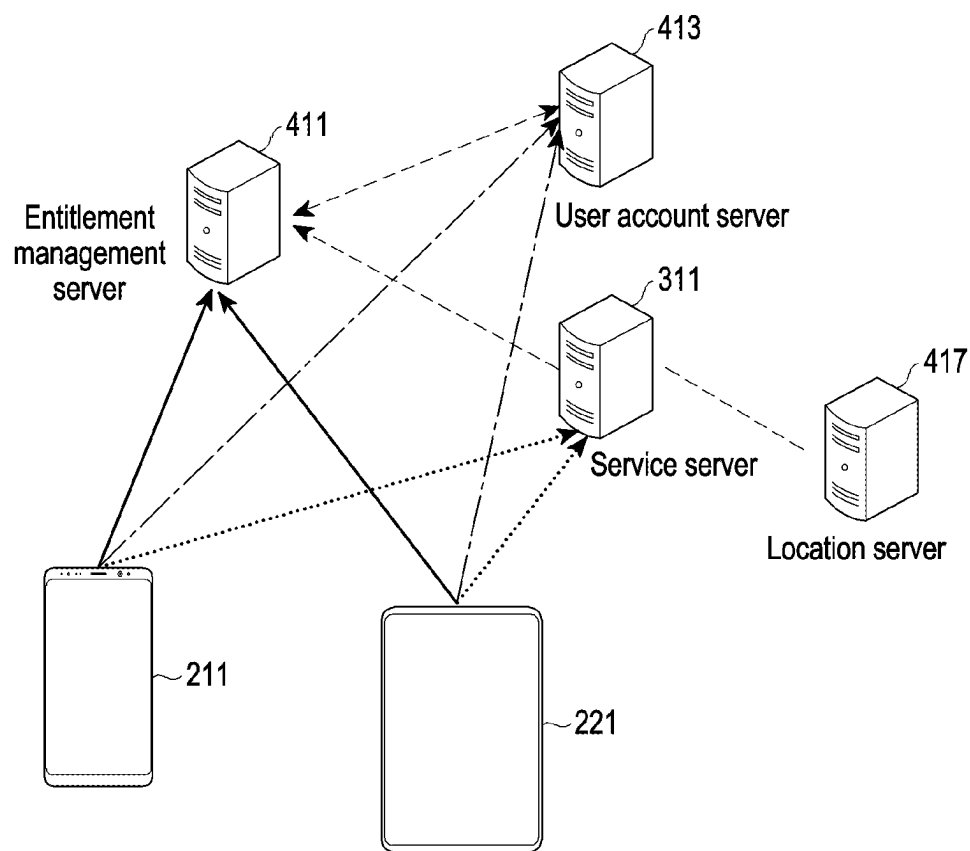


FIG. 4

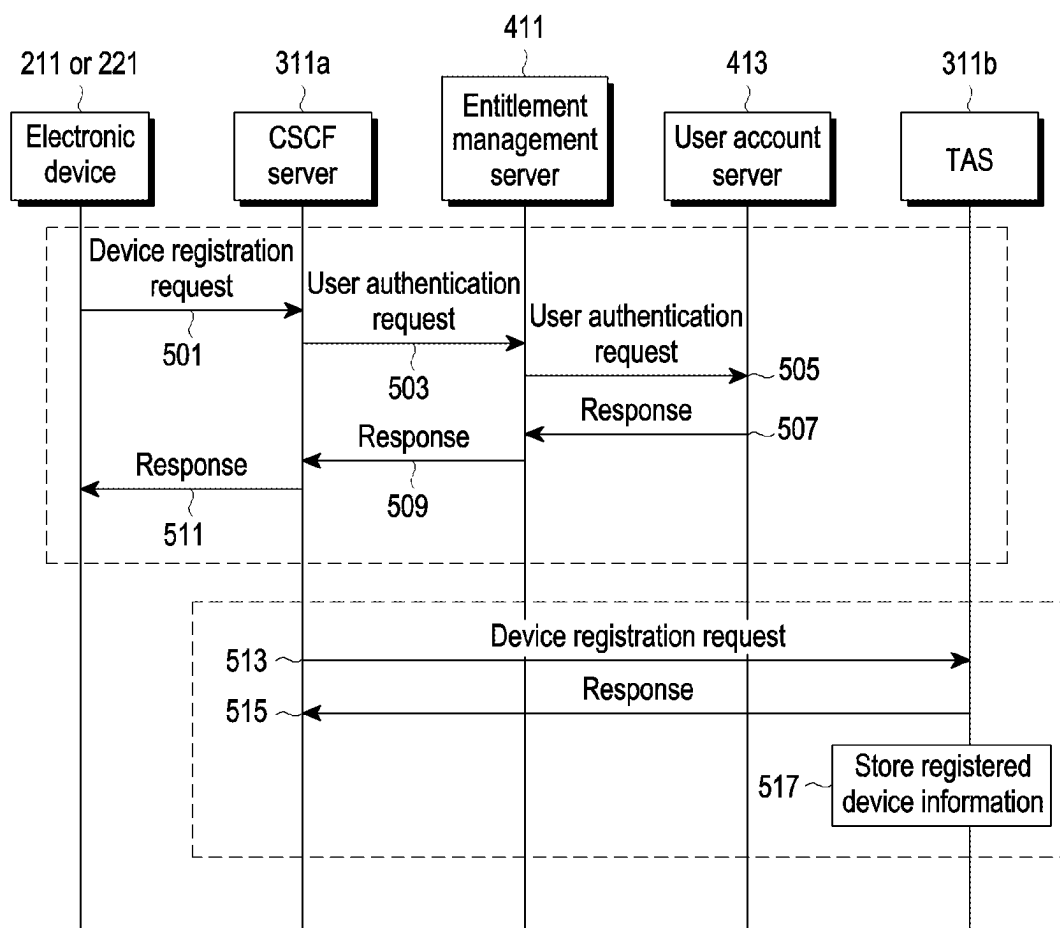


FIG. 5

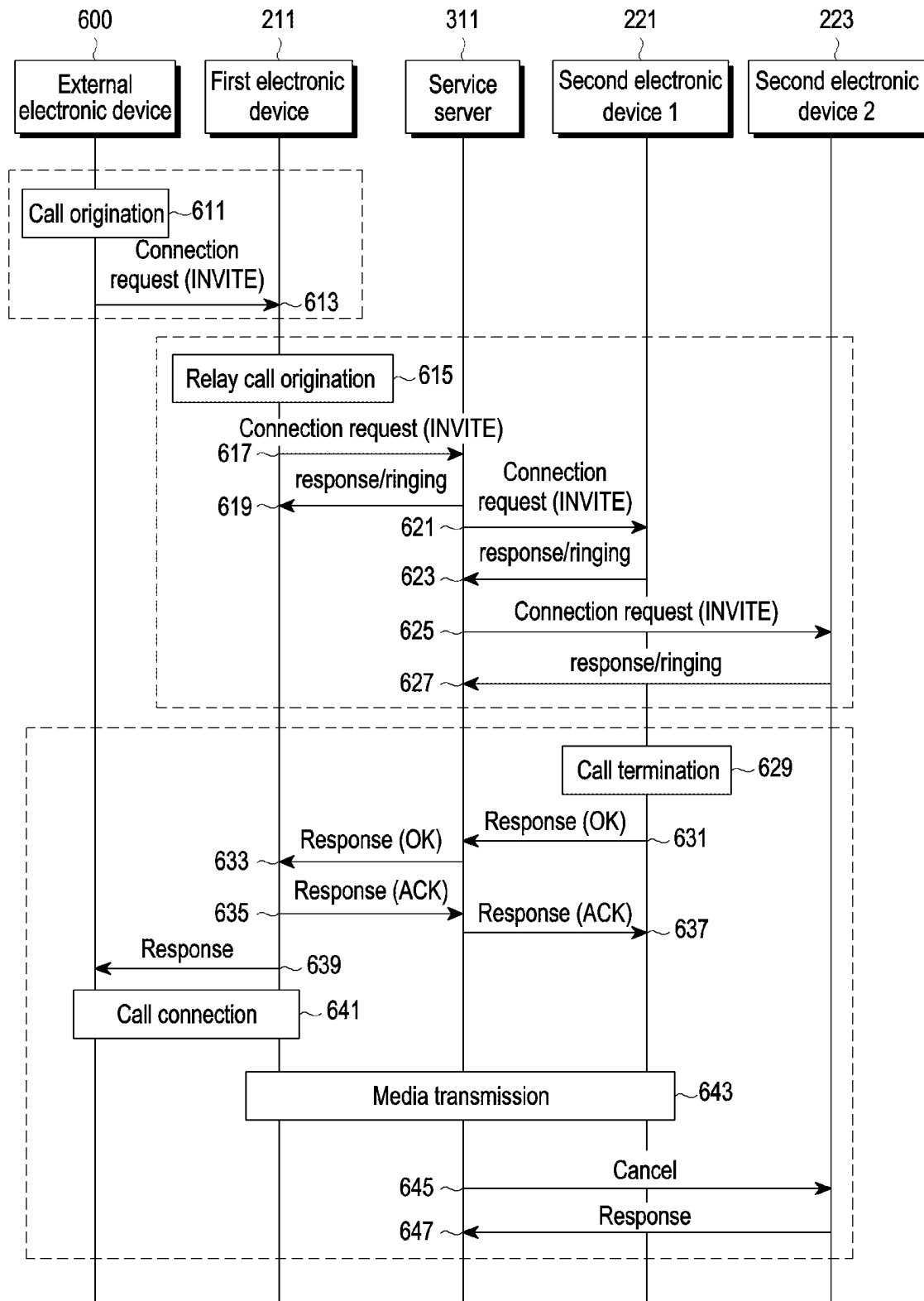


FIG. 6

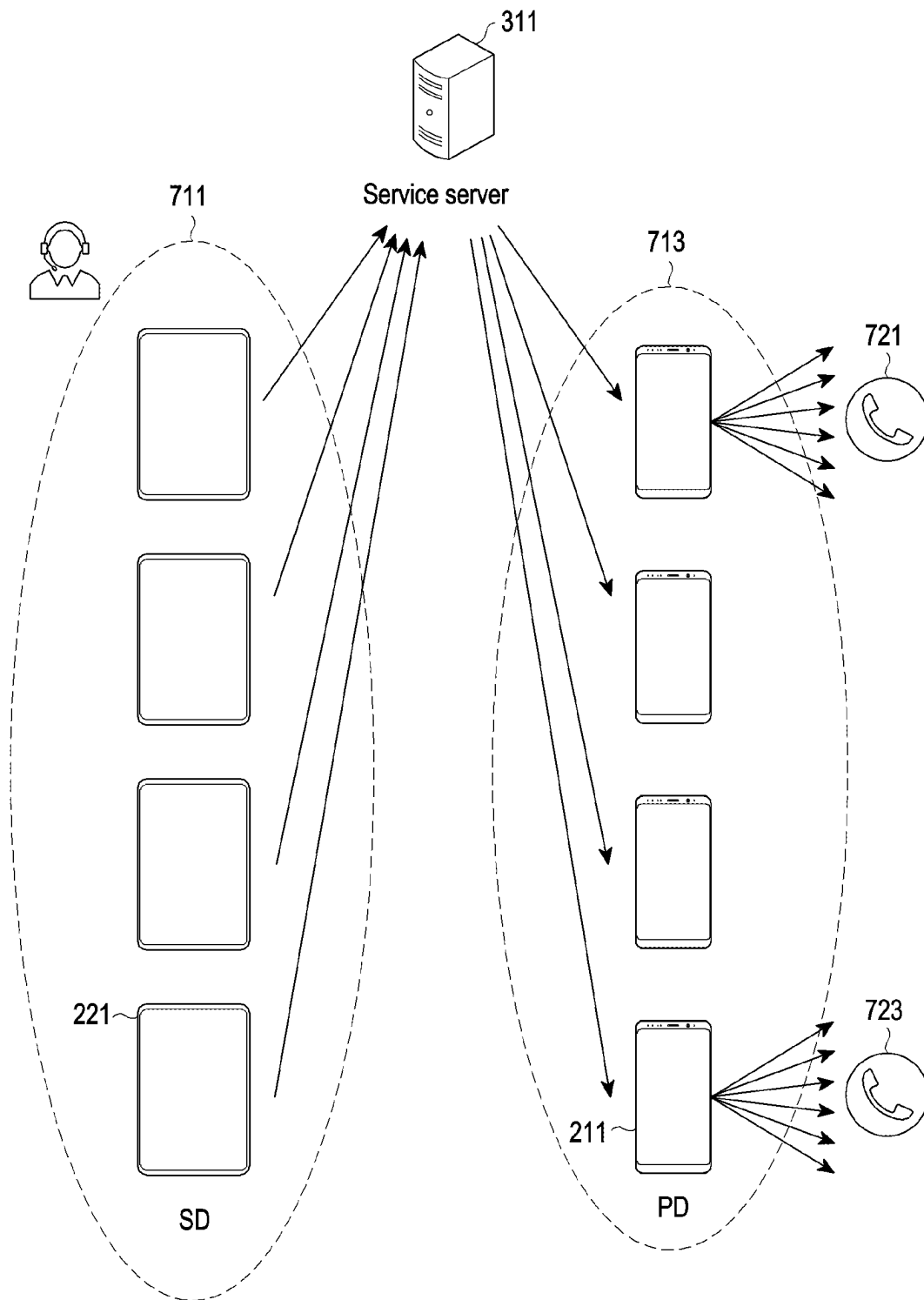


FIG. 7

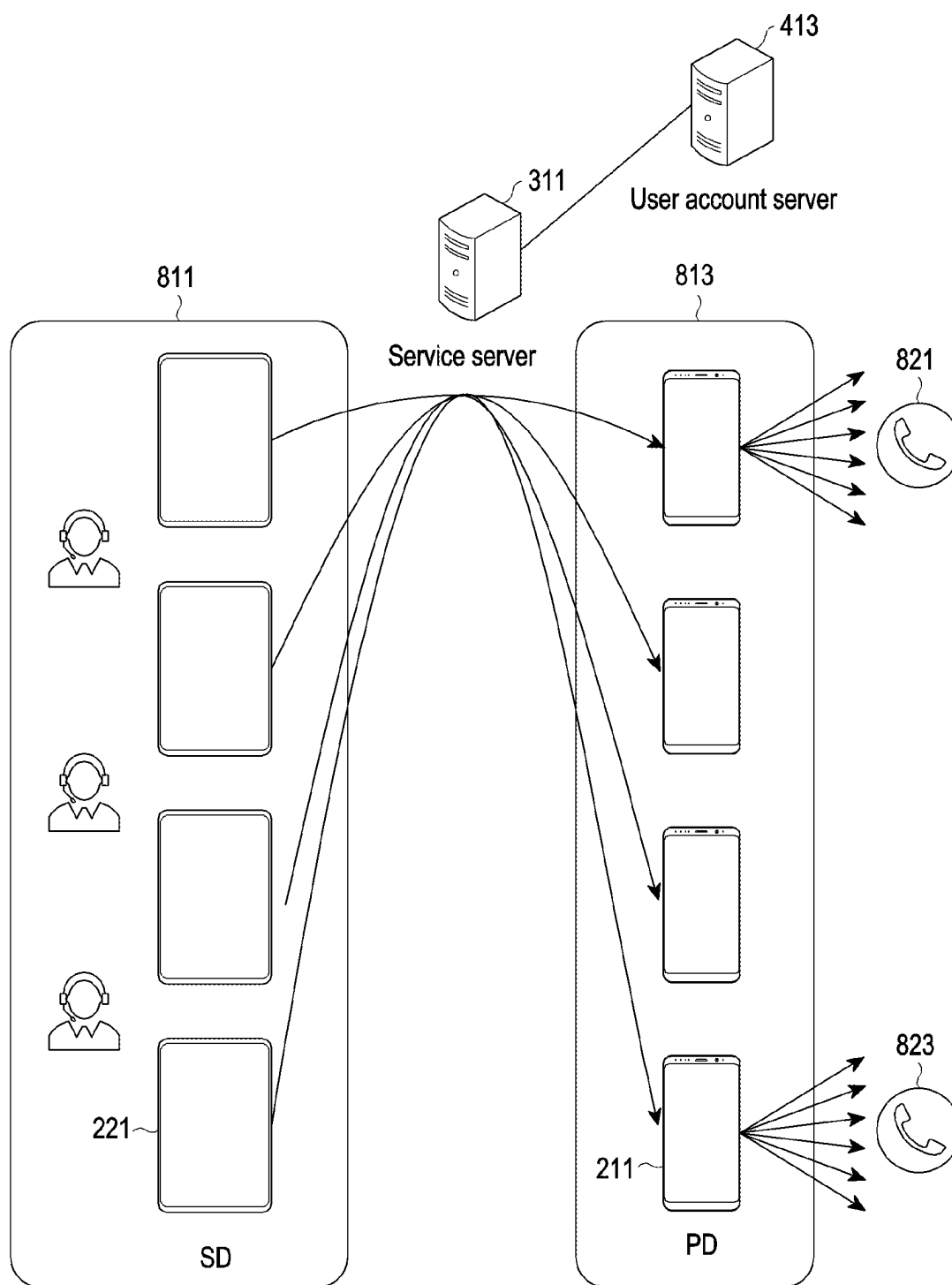


FIG. 8

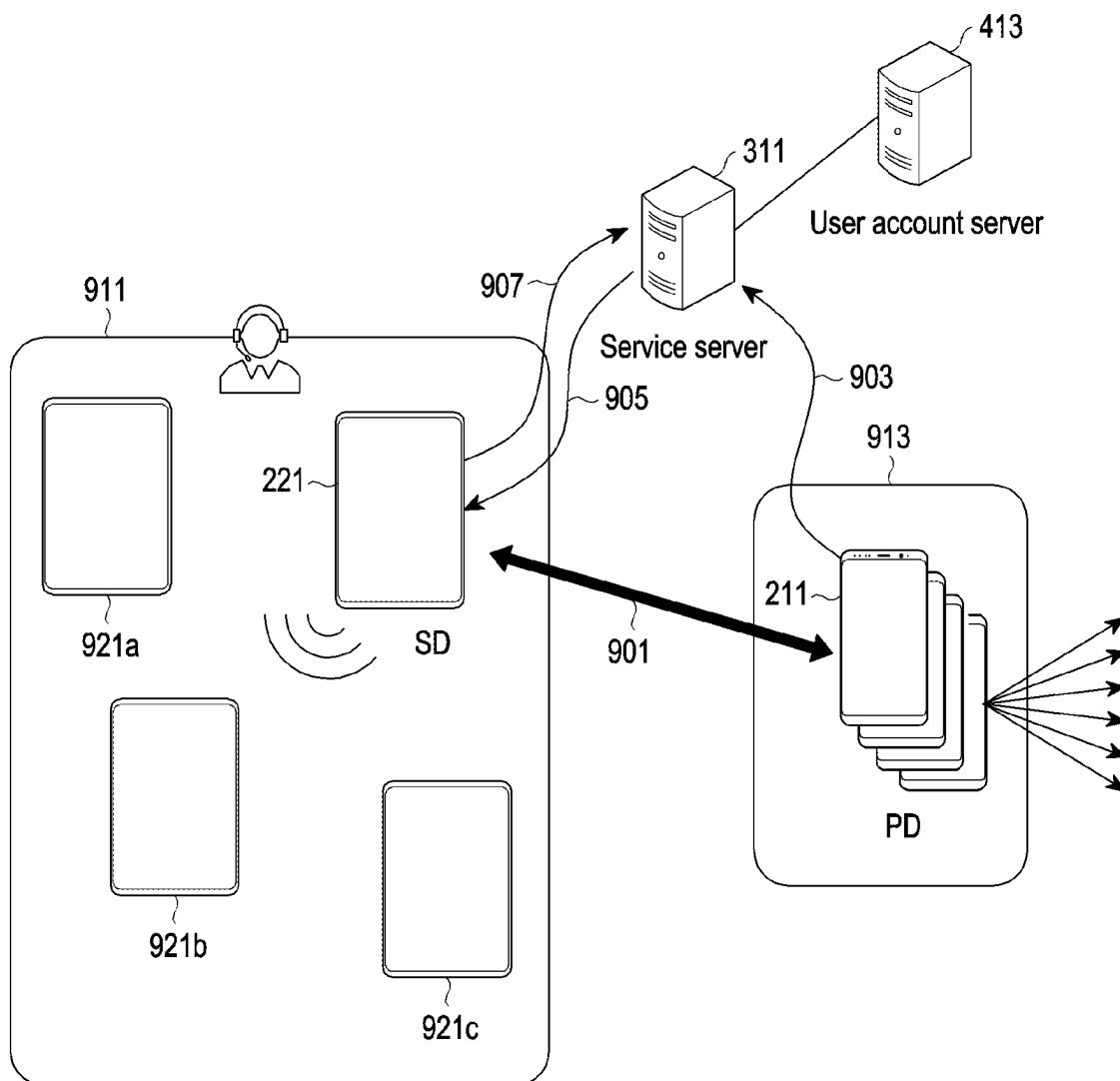


FIG. 9

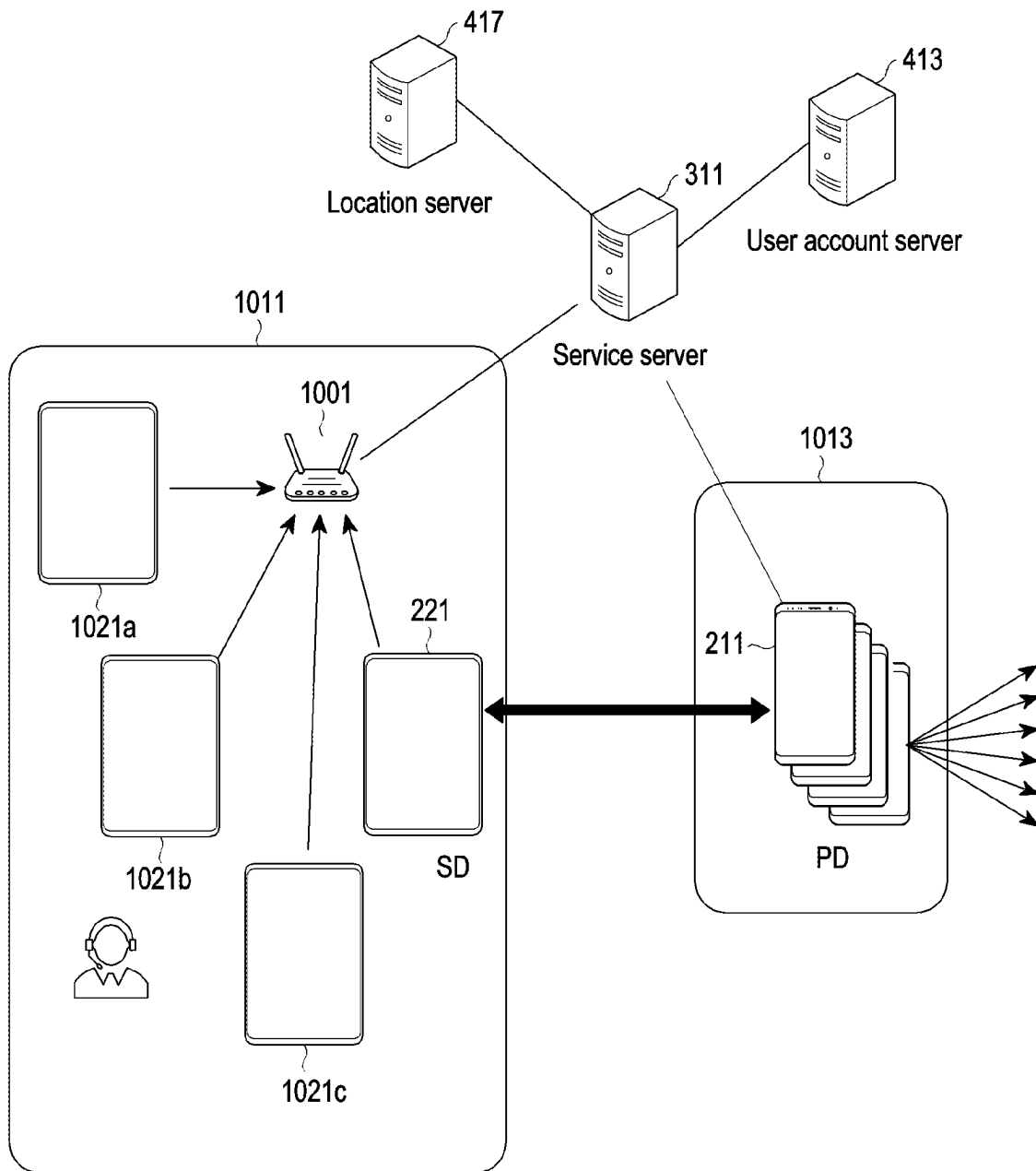


FIG. 10

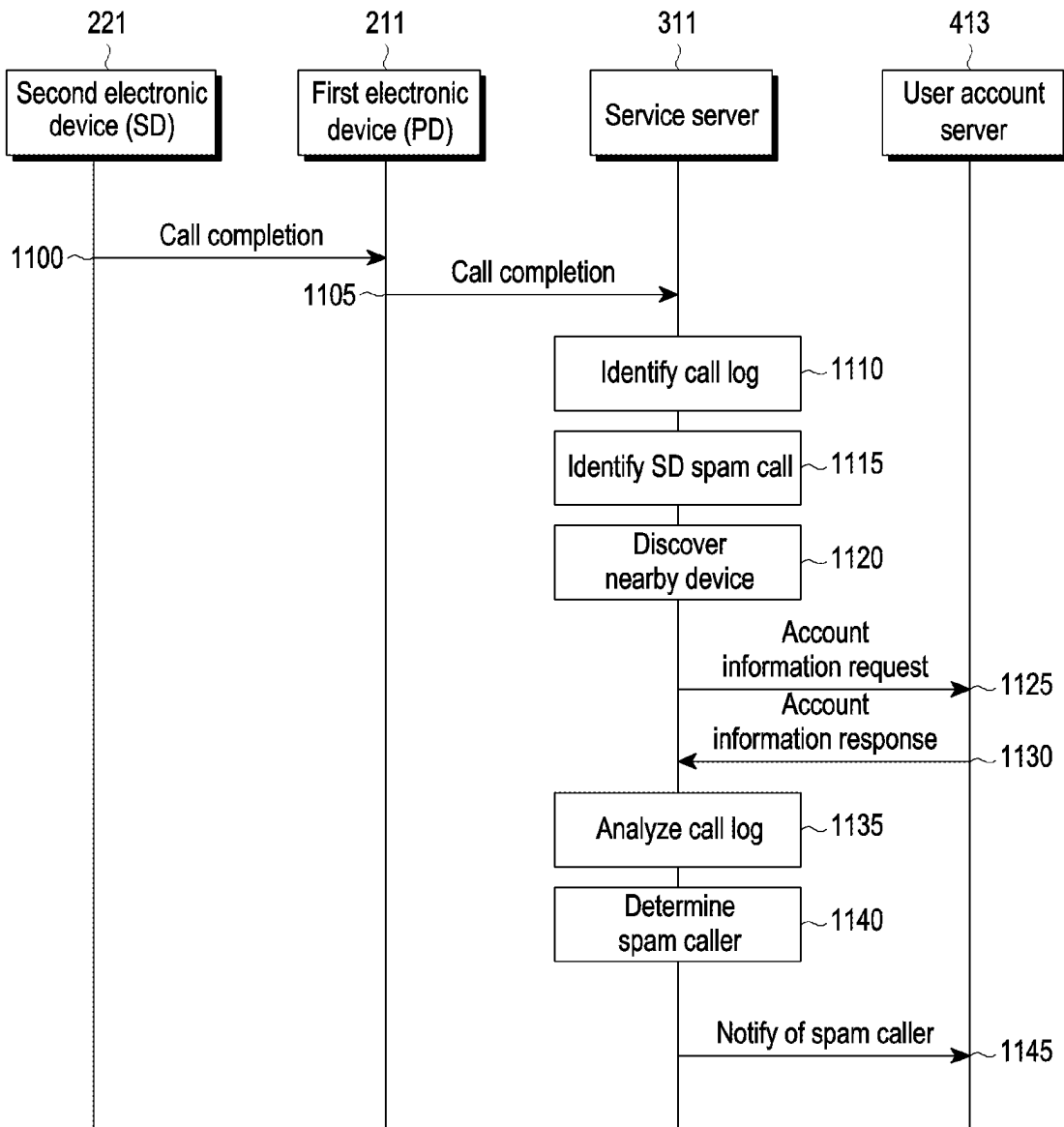


FIG. 11

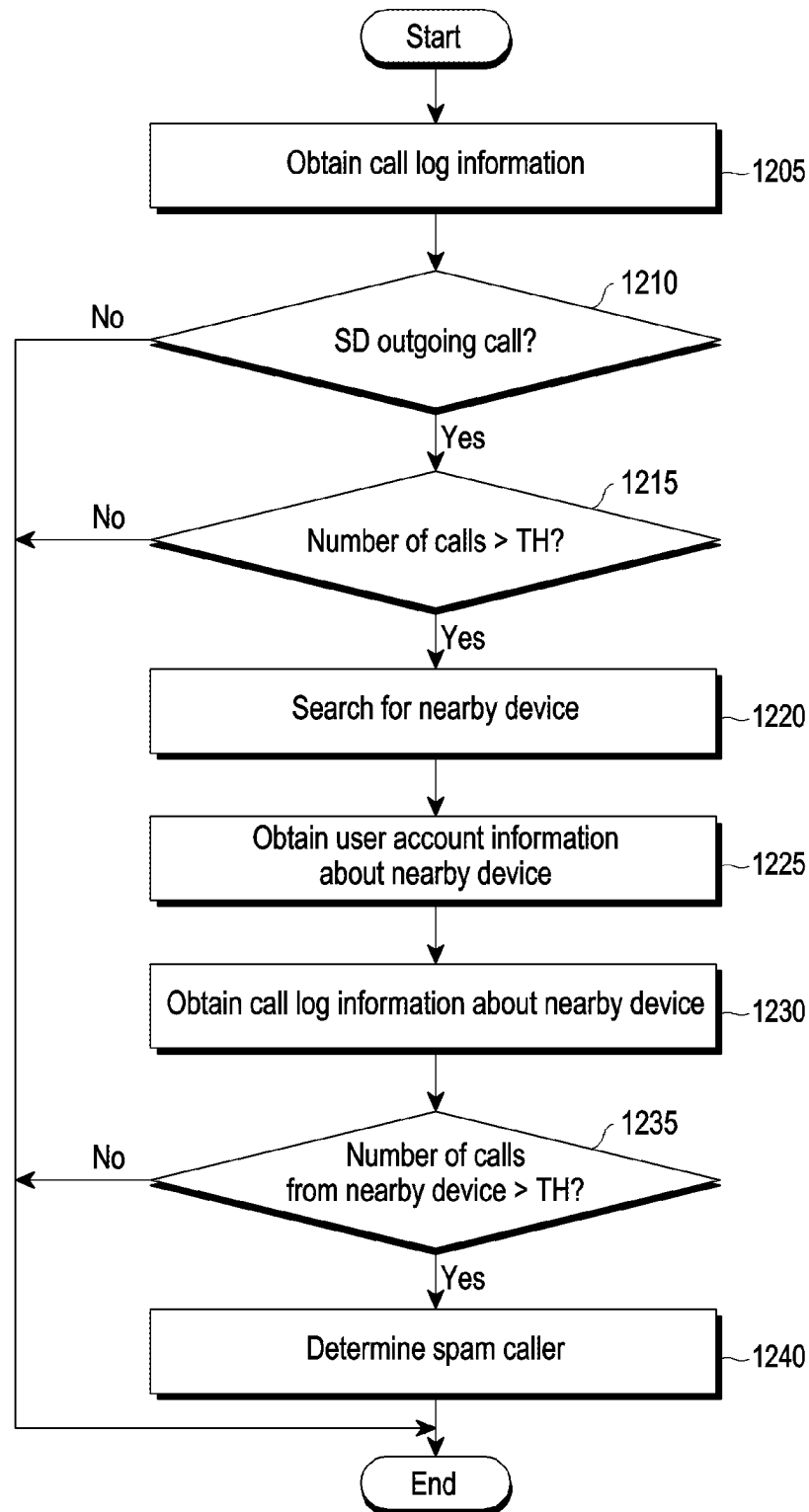


FIG. 12

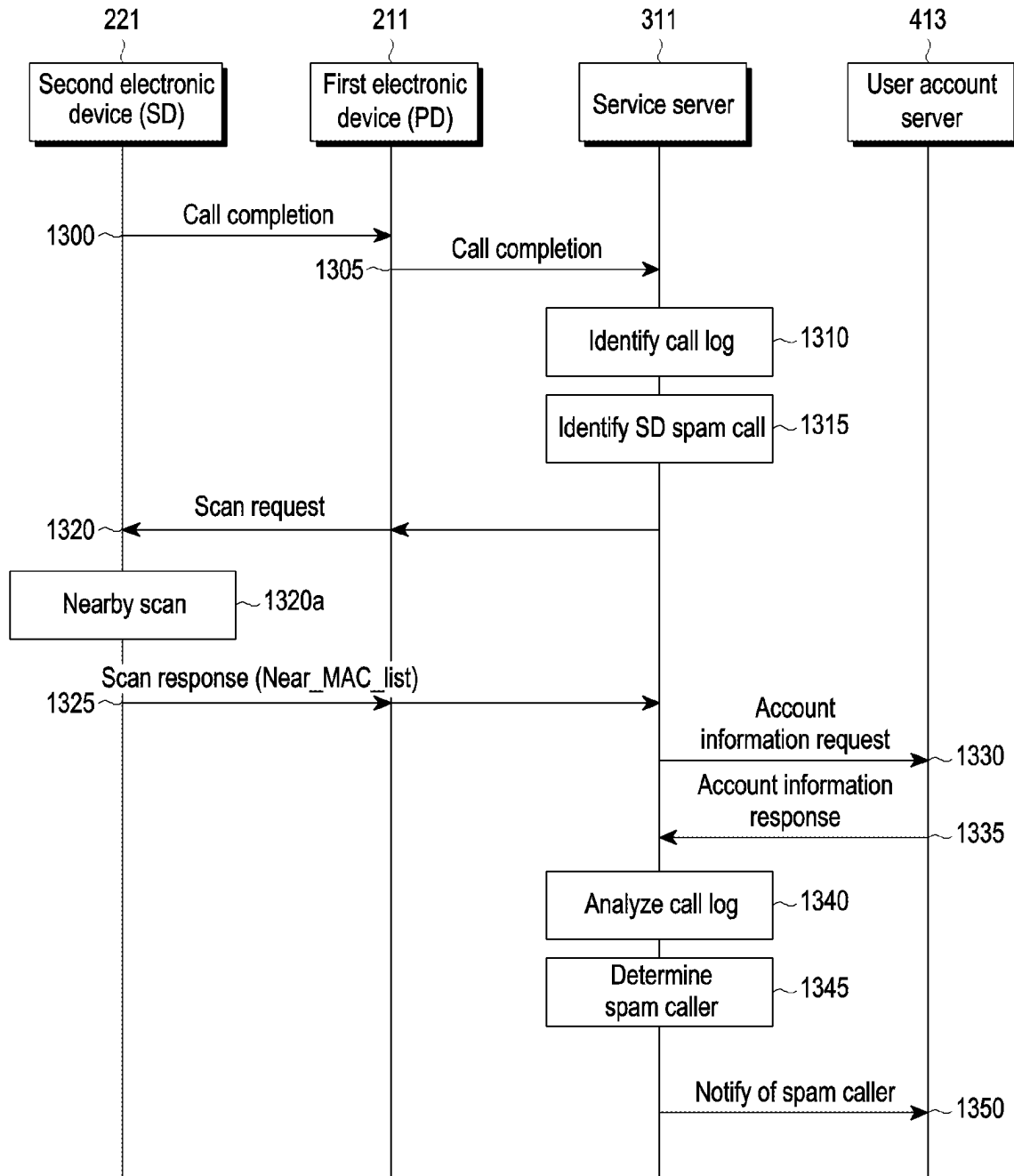


FIG. 13

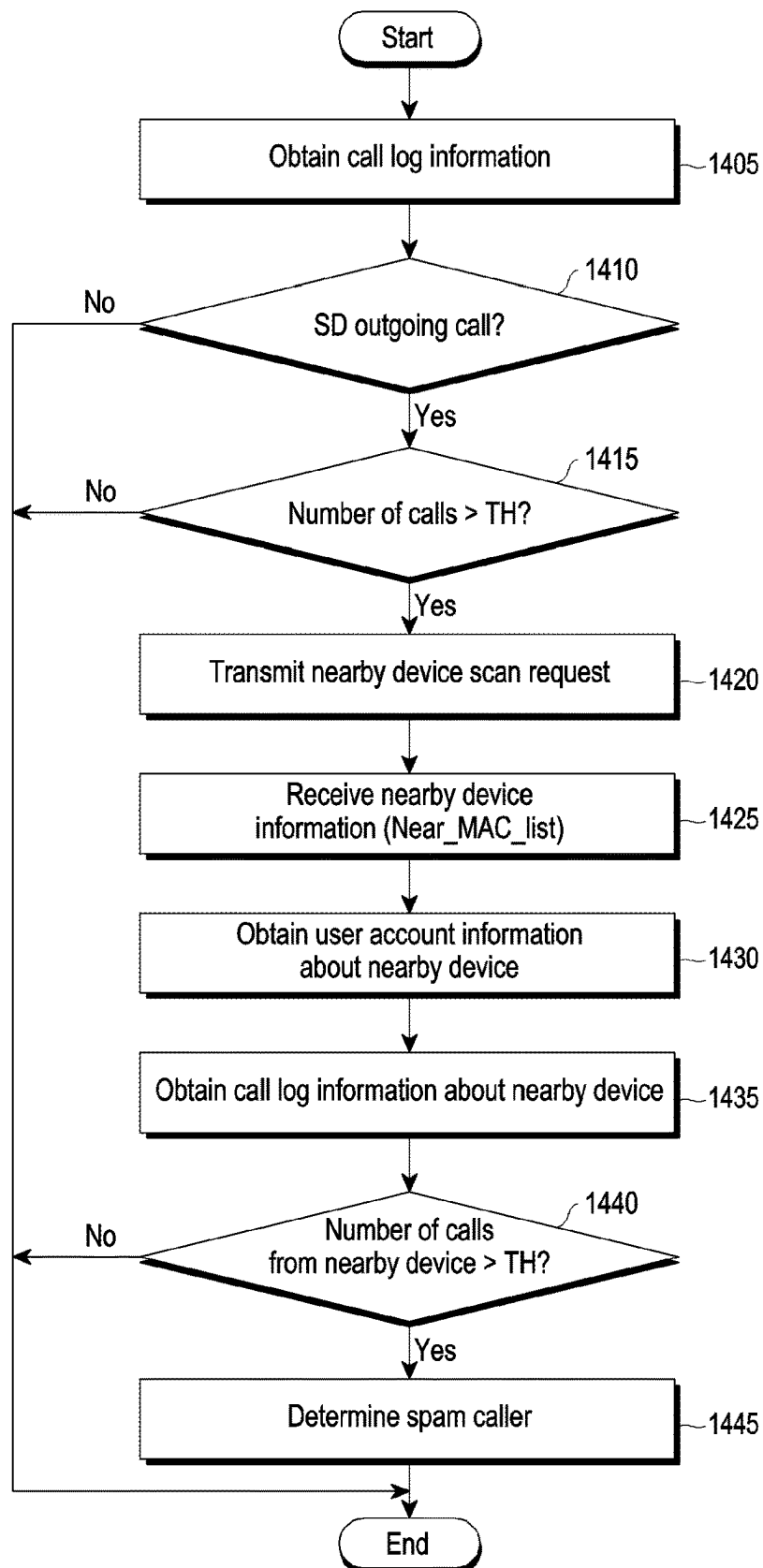


FIG. 14

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ELECTRONIC DEVICE FOR DETECTING SPAM CALL AND METHOD OF OPERATING THE SAME

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of International Application No. PCT/KR2023/001472, filed on Feb. 1, 2023, designating the United States, in the Korean Intellectual Property Receiving Office, and claiming priority to Korean patent application number 10-2022-0036287 filed on Mar. 23, 2022, the disclosures of which are all hereby incorporated by reference herein in their entireties.

BACKGROUND

Field

Certain example embodiments relate to an electronic device for detecting a spam call and/or a method of operating the same.

Description of Related Art

A communication business operator such as a wireless network, a voice call provider, or a messaging server may provide a messaging service or a voice call service over a packet-based network. Some users may abuse these services in an undesirable manner, for example, to transmit voice calls or a large number of messages (e.g., spam calls or spam messages) that receivers do not intend or want to receive.

Marketing strategies of transmitting calls or messages with promotional content such as an advertisement to electronic devices (e.g., smartphones) possessed by individuals have been actively carried out in recent years, and communication services of electronic devices may be misused even in crimes such as voice phishing.

SUMMARY

Accordingly, various techniques of detecting spam data (e.g., spam calls or spam messages) by monitoring calls or messaging are under study.

Some electronic devices may be connected to each other based on the same user account to provide a user with call and message continuity between the electronic devices.

For example, a smartphone and a tablet (or a wearable devices such as a smart watch) that share the same user account may be connected to their respective communication networks (e.g. a cellular network or wireless fidelity (WiFi)), and the smartphone may operate as a primary device with a subscriber identity module (SIM), whereas the tablet may operate as a secondary device without a SIM. A user may receive a call made to the smartphone on the tablet, and check a text message (e.g., a short messaging service (SMS) or a multimedia messaging service (MMS)) received in the smartphone on the tablet. In another example, the user may make a call or transmit a text message through the tablet, using the phone number of the smart phone.

A plurality of electronic devices (e.g., which may include a primary device and a secondary device) that are associated with each other based on a user account may be abused to transmit calls or messages for illegal or malicious purposes. In an example, a malicious user may place a primary device in one country and use a secondary device located in another country to make spam calls and/or transmit spam messages.

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Since the other party (or a message recipient) receives a call (or message) from the phone number (e.g., domestic number) of the primary device, the other party may not suspect or recognize the maliciousness of the caller (or message sender).

Blocking the call and message continuity function of electronic devices (e.g., secondary devices) located in a designated country to solve this problem may reduce user experiences and thus cause inconvenience to users.

In certain example embodiments, a spam caller who makes a spam call and/or transmits a spam message may be detected based on call log information of an electronic device and information about nearby devices.

In certain example embodiments, a user account used for a malicious action (e.g., spam calls and/or spam messaging) may be detected based on call log information of a plurality of electronic devices associated with each other based on the same user account.

In certain example embodiments, at least one of a plurality of electronic devices that provide call and message continuity may detect at least one external electronic device used for a malicious action by using call log information of nearby external electronic devices.

A server according to an example embodiment may include a communication circuit, and at least one processor. The at least one processor may be connected, directly or indirectly, to a first electronic device and a second electronic device, using the communication circuit. The at least one processor (comprising processing circuitry) may be configured to obtain first call log information including at least one outgoing call made using a call function of the first electronic device by the second electronic device. The at least one processor may be configured to determine whether the second electronic device satisfies a first condition for determining that the second electronic device is a spam caller, based on the first call log information. The at least one processor may be configured to search for one or more external electronic devices located around the second electronic device, based on the first condition being satisfied. The at least one processor may be configured to obtain second call log information about the one or more external electronic devices. The at least one processor may be configured to determine at least one of the one or more external electronic devices or the second electronic device as a spam caller, based on the second call log information.

An operation method of a server according to an example embodiment may include obtaining first call log information including at least one outgoing call made using a call function of a first electronic device by a second electronic device connected, directly or indirectly, to the server. The method may include determining whether the second electronic device satisfies a first condition for determining that the second electronic device is a spam caller, based on the first call log information. The method may include searching for one or more external electronic devices located around the second electronic device, based on the first condition being satisfied. The method may include obtaining second call log information about the one or more external electronic devices. The method may include determining at least one of the one or more external electronic devices or the second electronic device as a spam caller, based on the second call log information.

BRIEF DESCRIPTION OF THE DRAWINGS

The foregoing and other features of example embodiments will become more apparent from the following

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detailed description of embodiments when read in conjunction with the accompanying drawings. In the drawings, like reference numerals refer to like elements.

FIG. 1 is a block diagram illustrating an electronic device in a network environment according to various example embodiments.

FIG. 2 is a diagram illustrating an exemplary phone number-based communication group according to an example embodiment.

FIG. 3 is a block diagram illustrating a connection between a first electronic device and a second electronic device according to an example embodiment.

FIG. 4 is a block diagram illustrating a system for providing a call and message continuity function according to an example embodiment.

FIG. 5 is a diagram illustrating a signal flow for a device registration procedure for the call and message continuity function according to an example embodiment.

FIG. 6 is a diagram illustrating a signal flow for call termination based on the call and message continuity function according to an example embodiment.

FIG. 7 is a diagram illustrating a network environment according to an example embodiment.

FIG. 8 is a diagram illustrating a procedure of detecting a spam call according to an example embodiment.

FIG. 9 is a diagram illustrating an operation of searching for nearby devices according to an example embodiment.

FIG. 10 is a diagram illustrating an operation of searching for nearby devices, using location information according to an example embodiment.

FIG. 11 is a diagram illustrating a signal flow for an operation of detecting spam calls made by nearby devices according to an example embodiment.

FIG. 12 is a flowchart illustrating an operation of a service server for detecting spam calls made by nearby devices according to an example embodiment.

FIG. 13 is a diagram illustrating a signal flow for an operation of detecting a spam call, using nearby scan according to an example embodiment.

FIG. 14 is a flowchart illustrating an operation of a service server for detecting a spam call using nearby scan of an electronic device according to an example embodiment.

DETAILED DESCRIPTION

Certain example embodiments will be described below with reference to the attached drawings. Lest it should obscure the subject matter of the disclosure, a detailed description of a generally known function or structure of the disclosure will be avoided. Although the terms as described later are defined in consideration of functions in the disclosure, the terms may be changed according to the intention of a user or an operator, or customs. Therefore, the definitions should be made by the meaning of each term lying within.

The technical terms as used in the disclosure are provided merely to describe embodiments, not intended to limit the scope of the disclosure. Further, unless otherwise defined, the technical terms as used in the disclosure should be interpreted as the same meanings as generally understood by those skilled in the art, and should not be interpreted as excessively inclusive or excessively narrow meanings. In addition, when the technical terms used in the disclosure are incorrect ones that do not accurately express the subject of the disclosure, they should be understood by being replaced with technical terms that can be correctly understood by those skilled in the art. In addition, the general terms used in the disclosure should be interpreted as defined in dictionary

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ies or according to the context, and should not be interpreted as excessively narrow meanings.

Singular forms used in the disclosure include plural referents unless the context clearly dictates otherwise. In this application, the term “have”, “may have”, “include”, or “may include” should not be interpreted as necessarily including multiple components or steps described in the specification, and should be interpreted as excluding some of the components or steps or further including additional components or steps.

Further, the terms including first or second as used in the disclosure may be used to describe various components, and the components should not be limited by the terms. The terms are used only for the purpose of distinguishing one component from another. For example, a first component may be called a second component, and vice versa without departing from the scope of the disclosure.

When it is said that a component is “connected to” or “coupled to” another component, the component may be connected or coupled to the other component directly or with a third component in between. On the other hand, when it is said that a component is “directly connected to” or “directly coupled to” another component, it should be understood that there is no third component in between.

It should be noted that the attached drawings are presented merely to help understanding of the subject of the disclosure, and should not be construed as limiting the subject of the disclosure. The subject of the disclosure should be interpreted as encompassing all modifications, equivalents, and alternatives in addition to the attached drawings.

Although a mobile station will be described below with reference to the drawings, the mobile station may also be referred to as an electronic device, a terminal, a mobile equipment (ME), a user equipment (UE), a user terminal (UT), a subscriber station (SS), a wireless device, a handheld device, or an access terminal (AT). In addition, the mobile station may be a device equipped with a communication function, such as a portable phone, a personal digital assistant (PDA), a smartphone, a wireless modem, or a laptop computer.

FIG. 1 is a block diagram illustrating an electronic device 101 in a network environment 100 according to various embodiments. Referring to FIG. 1, the electronic device 101 in the network environment 100 may communicate with an electronic device 102 via a first network 198 (e.g., a short-range wireless communication network), or at least one of an electronic device 104 or a server 108 via a second network 199 (e.g., a long-range wireless communication network). According to an embodiment, the electronic device 101 may communicate with the electronic device 104 via the server 108. According to an embodiment, the electronic device 101 may include a processor 120, memory 130, an input module 150, a sound output module 155, a display module 160, an audio module 170, a sensor module 176, an interface 177, a connecting terminal 178, a haptic module 179, a camera module 180, a power management module 188, a battery 189, a communication module 190, a subscriber identification module (SIM) 196, or an antenna module 197. In some embodiments, at least one of the components (e.g., the connecting terminal 178) may be omitted from the electronic device 101, or one or more other components may be added in the electronic device 101. In some embodiments, some of the components (e.g., the sensor module 176, the camera module 180, or the antenna module 197) may be implemented as a single component (e.g., the display module 160).

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The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to an example embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

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The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic

device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

According to various embodiments, the antenna module **197** may form an mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-nearby communication scheme (e.g., a bus, general purpose input and output (GPIO), serial nearby interface (SPI), or mobile industry processor interface (MIPI)).

According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic

device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

Terms used to describe various embodiments may be as follows.

In various embodiments, a first electronic device (e.g., the electronic device **101** of FIG. **1**) may be a device capable of using a call function (and a message transmission and reception function) by connecting to a wireless communication network (e.g., the first network **198** (a short-range wireless communication network) or the second network **199** (a long-range wireless communication network) of FIG. **1**). For example, the first electronic device may be a device including a SIM. The first electronic device may be a primary device that provides call and message continuity to a second electronic device.

In various embodiments, the second electronic device (e.g., the electronic device **102** of FIG. **1**) may be an external electronic device connected to a network (e.g., the first network **198** (the short-range wireless communication network) or the second network **199** (the long-range wireless communication network) of FIG. **1**). For example, the second electronic device may not include a SIM. The second electronic device may be a secondary device that provides call and message continuity by using the call function of the first electronic device.

In an embodiment, aside from the use of the call and message continuity function through the first electronic device, the second electronic device may connect to a network (e.g., the first network **198** (the short-range wireless communication network) or the second network **199** (the long-range wireless communication network) of FIG. **1**) and use the call function. For example, the second electronic device (e.g., a tablet) may communicate with the first electronic device, using short-range wireless communication (e.g., wireless fidelity (WiFi)). In another example, the second electronic device (e.g., the tablet) may connect to a network (e.g., the second network **199** (the long-range wireless communication network) of FIG. **1**), using short-range wireless communication (e.g., WiFi).

A device with a SIM may be set as a primary device of a communication group. A device without a SIM may be set as a secondary device of the communication group.

For example, at least one of the first electronic device or the second electronic device may be a portable communication device (e.g., a smartphone), a wearable device (e.g., a smart watch), a portable computer (e.g., a tablet or a laptop), a computer, or a smart home appliance (e.g., a smart TV).

In an embodiment, a communication group may be a group of one first electronic device and at least one second electronic device connected, directly or indirectly, to the first electronic device. Communication between the second electronic device that is not connected to a network (e.g., the

second network **199** (the long-range wireless communication network) of FIG. **1**) and the network may be relayed by the first electronic device. The first electronic device and the second electronic device may be connected through, for example, a second network (e.g., the first network **198** (the short-range wireless communication network) of FIG. **1**).

In an embodiment, a phone number may be registered in one user account, and a communication group including a first electronic device and at least one second electronic device may be set in correspondence with the phone number. The phone number may be used as a group identifier (ID) identifying the communication group.

In an embodiment, the call and message continuity function may be a function or service of relaying communication between a first electronic device connected to a network (e.g., the second network **199** (the long-distance wireless communication network) of FIG. **1**) and a second electronic device that is not connected to the network (e.g., the second network **199** (the long-distance wireless communication network) of FIG. **1**). For example, the call and message continuity function may support a plurality of electronic devices (e.g., a primary device and at least one secondary device) to communicate using one phone number. The call and message continuity function may provide communication continuity between the electronic devices by connecting the at least one secondary device to the primary device. The call and message continuity function may synchronize calls and messages between a plurality of electronic devices connected to each other.

The call and message continuity function may include, for example, at least one of call forking, call transfer, message (e.g., short messaging service (SMS) message, multimedia messaging service (MMS) message, or social networking service (SNS) message) synchronization, or call log synchronization. A call may include at least one of a voice call, a video call, a media call (e.g., an IP multimedia subsystem (IMS) call), or a relay call. For example, call forking may include phone conversation relay.

Call forking may include a function of invoking a phone call at one phone number for a call request and ringing or vibrating multiple (e.g., two) electronic devices (e.g., a primary device and at least one secondary device). For example, when one of the electronic devices answers a call, a cancellation message may be transmitted to stop the call request for the other electronic device(s).

In various embodiments, the call and message continuity function may be a function of relaying a call between a first electronic device (e.g., a primary device) and an external electronic device (e.g., the electronic device **104** of FIG. **1**) that are connected to a network (e.g., the second network **199** (the long-range wireless communication network) of FIG. **1**) to a second electronic device (e.g., a secondary device) not connected to the network to provide the call between the second electronic device and the external electronic device.

For example, a second electronic device registered to a server (e.g., a service server) may receive a call which has been made from the external electronic device to the first electronic device through the network (e.g., the second network **199** (the long-distance wireless communication network) of FIG. **1**), or call the external electronic device through the first electronic device. In another example, the first electronic device may forward media (voice and/or video) received from the external electronic device to the second electronic device, and forward media (voice and/or video) received from the second electronic device to the external electronic device. The first electronic device and the

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second electronic device may be connected, directly or indirectly, to each other through a first network (e.g., the second network **199** (the long-range wireless communication network such as a cellular network) of FIG. 1) or a second network (e.g., the first network **198** (the short-range wireless communication network such as Wi-Fi) of FIG. 1).

Message synchronization may include a function of transmitting a message (e.g., SMS/MMS/SNSN message) transmitted to one phone number to a plurality of electronic devices (e.g., a primary device and at least one secondary device). Message synchronization may include a function of sharing or synchronizing message logs between the plurality of electronic devices (e.g., the primary device and the at least one secondary device). Message transmission and reception between an external electronic device and the first electronic device through a network may be relayed to the second electronic device. The second electronic device may transmit and receive a message through the first electronic device.

Call log synchronization may include a function of sharing or synchronizing their respective call logs between a plurality of electronic devices (e.g., a primary device and at least one secondary device). A call log stored in a first electronic device may be transmitted to a second electronic device, and the second electronic device may check the call log. The second electronic device may transmit or receive a call through the first electronic device.

In various embodiments, the call and message continuity function may be provided to a plurality of electronic devices (e.g., a primary device and at least one secondary device) registered in one communication group. For example, the plurality of electronic devices in the communication group may communicate with an external electronic device based on the same phone number. For example, at least one phone number may be registered in a user account, and at least one communication group corresponding to each phone number may be set. Each communication group may include a first electronic device (e.g., primary device) and at least one second electronic device (e.g., secondary device).

Communication group setting may be at least one of group creation, group addition, device addition, group update, group deletion, or device deletion.

For example, in the case where the user purchases or activates the first electronic device and then intends to use a communication relay function in the first electronic device, when the communication relay function is initially activated in the first electronic device, group addition (creation/addition of a communication group) may be performed.

For example, in the case where the user purchases or activates the second electronic device and then connects the second electronic device to the first electronic device to use the communication relay function, when the communication relay function is initially activated in the second electronic device, device addition (addition of a device in a communication group) may be performed.

For example, in the case where the user moves a SIM (e.g., a SIM card) from an existing electronic device (a previous primary device) to a new electronic device, and sets the new electronic device as a new primary device, when the user intends to change the name of a communication group or the activation state of the communication relay function of the communication group, group update (communication group update) may be performed.

For example, group deletion (deletion of a communication group) may be performed, when the user leaves the

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account or wants to withdraw from the communication relay function of a communication group (e.g., when the user changes a phone number).

For example, device deletion (deletion of a device within a communication group) may be performed, when the user deletes a device belonging to one communication group from the communication group (e.g., when the first electronic device connected to the second electronic device is changed).

FIG. 2 is a diagram illustrating an exemplary phone number-based communication group according to an embodiment.

Referring to FIG. 2, a phone number corresponding to a SIM **211a** may be registered in a user account **201**, and a communication group **231** may be set in correspondence with the phone number. The communication group **231** may include a first electronic device **211** and at least one second electronic device (e.g., second electronic device **1 221** or second electronic device **2 223**). The call and message continuity function based on the same phone number may be provided within the communication group **231**.

The first electronic device **211** may include a SIM **211a** (e.g., a SIM card) corresponding to the phone number. The SIM **211a** may be the SIM **196** of FIG. 1.

For example, a user may register second electronic device **1 221** (e.g., a tablet or a watch) and/or second electronic device **2 223** (e.g., a tablet or a watch) in the communication group **231** set based on the phone number registered in the user account **201**. According to an embodiment, second electronic device **1 221** and/or second electronic device **2 223** may not include any SIM. In an embodiment, the first electronic device **211** may be located in the same space as or a space (e.g., a room, a building, or a country) separated from second electronic device **1 221** and/or second electronic device **2 223**. The first electronic device **211** may be connected to a first network (e.g., the first network **198**) or a second network (e.g., the second network **199**). Likewise, second electronic device **1 221** and/or second electronic device **2 223** may be connected, directly or indirectly, to the first network (e.g., the first network **198**) or the second network (e.g., the second network **199**).

The user may make a call using at least one of second electronic device **1 221** or second electronic device **2 223**, and the call may be transferred to a peer electronic device by the first electronic device **211**. An incoming call received from the peer electronic device may be transferred to second electronic device **1 221** or second electronic device **2 223** by the first electronic device **211**, and when any of the first electronic device **211**, second electronic device **1 221**, and second electronic device **2 223** answers the incoming call, an incoming call notification (e.g., ringing) may stop in the other devices.

FIG. 3 is a block diagram illustrating a connection between a first electronic device and a second electronic device according to an embodiment. In certain example embodiments, the first electronic device **211** or the second electronic device **221** may include all or part of the electronic device **101** illustrated in FIG. 1.

Referring to FIG. 3, the first electronic device **211** may include the SIM **211a** and directly connect to a first network (e.g., the second network **199** (the long-range wireless communication network) of FIG. 1) without communication relay. In an embodiment, the first electronic device **211** and the second electronic device **221** may communicate with a service server **311** through the first network (e.g., the second network **199** (the long-range wireless communication network) of FIG. 1) or a second network (e.g., the first network

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198 (the short-range wireless communication network) of FIG. 1). For example, the first electronic device 211 may communicate with the service server 311 through the first network (e.g., the second network 199 (the long-range wireless communication network) of FIG. 1) or the second network (e.g., the first network 198 (the short-range wireless communication network) of FIG. 1), and the second electronic device 221 may communicate with the service server 311 through the second network (e.g., the first network 198 (the short-range wireless communication network) of FIG. 1). In an embodiment, the second electronic device 221 may be connected to the first electronic device 211 using WiFi or Bluetooth, and communicate with the service server 311 through the first electronic device 211.

The service server 311 may include a processor 313 (e.g., the processor 120 of FIG. 1, comprising processing circuitry), a communication circuit 315 (e.g., the communication module 190 of FIG. 1), and memory 317 (e.g., the memory 130 of FIG. 1).

The first electronic device 211 may include a display 213 (e.g., the display module 160 of FIG. 1 comprising at least one display), a processor 215 (e.g., the processor 120 of FIG. 1), a communication circuit 217 (e.g., the communication module 190 of FIG. 1 comprising communication circuitry), memory 218 (e.g., the memory 130 of FIG. 1), and the SIM 211a (e.g., the SIM 196 of FIG. 1).

The second electronic device 221 may include a display 223 (e.g., the display module 160 of FIG. 1), a processor 225 (e.g., the processor 120 of FIG. 1), a communication circuit 227 (e.g., the communication module 190 of FIG. 1), and memory 229 (e.g., the memory 130 of FIG. 1).

The first electronic device 211 may autonomously connect to the first network (e.g., the second network 199 (the long-range wireless communication network) of FIG. 1) using the SIM 211a (e.g., the SIM 196 of FIG. 1).

In an embodiment, the second electronic device 221 may be configured to be connected, directly or indirectly, to the first electronic device 211 through the service server 311 to use the call function of the first electronic device 211. To this end, the second electronic device 221 may communicate with the service server 311 through the first network or the second network. In an embodiment, the second electronic device 221 may not include or support a SIM, and may connect to the service server 311 through the second network or through the first electronic device 211. In an embodiment, the second electronic device 221 may include its own SIM (e.g., the SIM 196 of FIG. 1), autonomously connect to the first network, using the SIM, and communicate with the service server 311 through the first network.

Without a SIM, the second electronic device 221 may use the communication and message continuity function based on short-range wireless communication (e.g., WiFi or Bluetooth). Even when the first electronic device 211 and the second electronic device 221 are spaced apart from each other by more than the coverage of short-range wireless communication, the second electronic device 221 may be connected to the service server 311 via a short-range wireless communication node (e.g., a wireless access point) to use the call and message continuity function of the first electronic device 211.

In an embodiment, the first electronic device 211 and the second electronic device 221 may be connected, directly or indirectly, to the same second network (e.g., the first network 198 (the short-range wireless communication network) of FIG. 1). The second electronic device 221 may be connected to the first electronic device 211 through the second network. In an embodiment, the second electronic device 221 may be

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connected to a second network different from that of the first electronic device 211. The second electronic device 221 may communicate with the service server 311 through the second network.

In an embodiment, the service server 311 (e.g., the processor 313) may receive a connection request for an incoming call corresponding to a phone number stored in the SIM 211a of the first electronic device 211 from the first electronic device 211 or from a service server connected, directly or indirectly, to a call peer (e.g., an external electronic device) through the communication circuit 315. The service server 311 (e.g., the processor 313, comprising processing circuitry) may identify the second electronic device 221 registered to provide the call and message continuity function for the first electronic device 211, in response to the connection request. The service server 311 (e.g., the processor 313) may transfer the connection request for the incoming call to the first electronic device 211 and/or the second electronic device 221, and accordingly, one or both of the first electronic device 211 and/or the second electronic device 221 may ring or vibrate for the incoming call. In an embodiment, one or both of the first electronic device 211 and/or the second electronic device 221 may display information (e.g., a caller number) corresponding to the incoming call on the display module 213 and/or the display module 223.

In an embodiment, upon receipt of an incoming call response corresponding to the incoming call from the second electronic device 221 earlier than an incoming call response corresponding to the incoming call from the first electronic device 211, the service server 311 may connect the incoming call to the second electronic device 221. In an embodiment, voice, video or media corresponding to the incoming call may be transferred from the first electronic device 211 to the second electronic device 221 through the service server 311, or (e.g., when the first electronic device 211 and the second electronic device 221 belong to the same network), from the first electronic device 211 to the second electronic device 221. In an embodiment, voice, video or media corresponding to the incoming call may be transferred from the second electronic device 221 to the first electronic device 211 through the service server 311 or from the second electronic device 221 to the first electronic device 211.

The first electronic device 211 may transfer the voice, video, or media to the call peer (e.g., the external electronic device) corresponding to the incoming call. In an embodiment, voice, video, or media generated from the first electronic device 211 or the second electronic device 221 may be transferred from the service server 311 to the call peer (e.g., the external electronic device) corresponding to the incoming call.

In an embodiment, the second electronic device 221 may use the call function (e.g., call origination) of the first electronic device 211 through the service server 311. An outgoing call request from the second electronic device 221 may be transferred to the first electronic device 211 through the service server 311, and the first electronic device 211 may transfer the outgoing call request to the call peer (recipient). Upon receipt of an outgoing call response corresponding to the outgoing call request from the call peer, the first electronic device 211 may transfer the outgoing call response to the second electronic device 221 through the service server 311 or directly. Media (voice and/or video) generated in a call corresponding to the outgoing call request may be transferred from the first electronic device 211 to the

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second electronic device **221**, or from the first electronic device **211** to the second electronic device **221** through the service server **311**.

FIG. **4** is a block diagram illustrating a system for providing the call and message continuity function according to an embodiment.

Referring to FIG. **4**, an entitlement management server **411** may store and manage information (e.g., user information, account information, a phone number, device information, and/or group information) about each user's communication group, and provide information about the communication group to the service server **311**, upon request. The service server **311** may access the entitlement management server **411** and identify information about the communication group of the first electronic device **211** or the second electronic device **221**, and set the communication group (e.g., perform group creation, group addition, group deletion, group update, device addition, or device deletion).

In an embodiment, a user account server **413** may store and manage information about a user account (e.g., a user ID or an account name), and process user authentication. In an embodiment, the user authentication may be performed in conjunction with the entitlement management server **411**. According to an embodiment, upon log-in on a user account, the user account server **413** may generate temporary authentication information. The entitlement management server **411** may request user authentication from the user account server **413**, using a user ID and temporary authentication information received from the first electronic device **211** or the second electronic device **221** through the service server **311**. The user account server **413** may process the user authentication request from the entitlement management server **411** and provide a response (accept or reject) to the user authentication request.

In an embodiment, the service server **311** may include one or more servers to provide the call and message continuity function. In an embodiment, the service server **311** may include a call session control function (CSCF) server, and/or a telephony application server (TAS). In an embodiment, the service server **311** may include an IMS server to support media calls. In an embodiment, the service server **311** may include a message relay server supporting a message relay function. In an embodiment, the message relay server may provide a message synchronization function.

For example, the CSCF server may operate as at least one of a proxy call session control function (P-CSCF), an interrogating call session control function (I-CSCF), or a serving call session control function (S-CSCF) server, which forms an IMS network. The P-CSCF server may operate as a contact point with the electronic device **211** or **221** in the IMS network, and may proxy session initiation protocol (SIP) messages. The P-CSCF server may provide functions such as security and/or compression in conjunction with the electronic device **211** or **221**. The I-CSCF server may be responsible for routing SIP messages to another network node (e.g., another CSCF or the electronic device **211** or **221**) in the IMS network to provide the call and message continuity function.

In an embodiment, the S-CSCF server may manage a session state for SIP messages. The S-CSCF server may perform call control required to provide a subscriber-requested service through routing to the TAS. The S-CSCF server may serve as an SIP register to process a device registration request from the electronic device **211** or **221**, and manage a device registration state.

In an embodiment, the TAS may provide an IMS-based multimedia communication service. The TAS may provide

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the call and message continuity function according to certain example embodiments by supporting an IMS-based media call.

In an embodiment, the service server **311** may identify communication group information from the entitlement management server **411**. In an embodiment, when the first electronic device **211** or the second electronic device **221** requests device registration from the service server **311**, for the call and message continuity function, the service server **311** may identify whether the device **211** or **221** which has transmitted the request or the communication group is registered in the entitlement management server **411**, and then perform authentication for the device **211** or **221** through the user account server **413**. In an embodiment, when the first electronic device **211** or the second electronic device **221** transmits a request (e.g., hyper-text transfer protocol (HTTP) request) for a synchronization service to the service server **311**, the service server **311** may identify whether the device **211** or **221** which has transmitted the request or the communication group is registered in the entitlement management server **411**, and then perform authentication for the device **211** or **221** through the user account server **413**.

In an embodiment, the service server **311** may communicate with a location server **417** that tracks and manages the location of the first electronic device **211** and/or the second electronic device **221**. The location server **417** may collect and store the location of the first electronic device **211** and/or the second electronic device **221**. In an embodiment, the location may include a cell ID, access point information, and/or a global positioning system (GPS) location (a latitude and a longitude).

In an embodiment, the entitlement management server **411** may obtain and identify device configuration information on a communication group basis. The entitlement management server **411** may identify information (e.g., the address(es) of server(s) required to provide a communication relay function) related to a service subscribed to by the user from the service server **311**.

In an embodiment, the first electronic device **211** or the second electronic device **221** may identify device configuration information from the service server **311** related to the call and message continuity function, and use the call and message continuity function (e.g., call forking or call log synchronization) in conjunction with the service server **311** based on the device configuration information.

In an embodiment, the first electronic device **211** or the second electronic device **221** may identify communication group information about the user from the entitlement management server **411**, and use the call and message continuity function (e.g., call forking or call log synchronization) in conjunction with the service server **311** based on the communication group information.

FIG. **5** is a diagram illustrating a signal flow for a device registration procedure for the call and message continuity function according to an embodiment. According to embodiments, at least one of the following operations may be skipped, changed, or reordered.

Referring to FIG. **5**, operations **501** to **511** may be a user authentication procedure.

In operation **501**, the electronic device **211** or **221** may transmit a device registration request to the service server **311** (e.g., a CSCF server **311a**). The device registration request may include at least one of a user ID, a group ID, a device ID, or temporary authentication information.

In operation **503**, the CSCF server **311a** may transmit a user authentication request to the entitlement management

server **411** to identify whether the device registration request is for an authorized user. The user authentication request may include at least one of the user ID, the group ID, the device ID, or the temporary authentication information obtained from the device registration request.

In operation **505**, the entitlement management server **411** may transmit a user authentication request to the user account server **413**. The user authentication request may include at least one of the user ID or the temporary authentication information.

In operation **507**, the user account server **413** may transmit a response based on an authentication result (success or failure) to the entitlement management server **411**.

In operations **509** and **511**, the response based on the authentication result may be transferred to the CSCF server **311a** and the electronic device **211** or **221**.

Operations **513** to **517** may be a device registration procedure.

In operation **513**, the CSCF server **311a** may transmit a device registration request to the service server **311** (e.g., a TAS **311b**). In operation **515**, the TAS **311b** may perform device registration according to the device registration request, and transmit a response (registration success or registration failure) to the CSCF server **311a**. In an embodiment, the TAS **311b** may refer to the result of the user authentication in operation **509**, for the device registration.

In operation **517**, the TAS **311b** may register and store device information (e.g., a model name, a model number, a serial number, an International Mobile Equipment Identity (IMEI), and/or a medium control access (MAC) address) about the electronic device **211** or **221**, corresponding to the device registration request.

FIG. 6 is a diagram illustrating a signal flow for call termination based on the call and message continuity function according to an embodiment. In the illustrated case, a call originating from a peer electronic device **600** is received by second electronic device **1 221**, by way of example.

Referring to FIG. 6, operations **611** and **613** may be a call origination procedure.

In operation **611**, the peer electronic device **600** (e.g., a caller) may originate a call by designating a called phone number according to a user input. The call may be, for example, a voice over long term evolution (VoLTE) call. The called phone number may be, for example, a phone number of the first electronic device **211**.

In operation **613**, a service server (not shown) of the peer electronic device **600** may transmit a connection request (e.g., an SIP INVITE message) for the call made by the peer electronic device **600** to the first electronic device **211** corresponding to the called phone number. The connection request may include the called phone number.

Operations **615** to **627** may be a call termination request procedure.

In operation **615**, the first electronic device **211** may originate a relay call corresponding to the call. In an embodiment, the first electronic device **211** may identify that the call and message continuity function is activated by a user setting, and determine to originate the relay call. In an embodiment, the first electronic device **211** may identify at least one secondary device (e.g., second electronic device **1 221** and second electronic device **2 223**) registered for the call and message continuity function.

In operation **617**, the first electronic device **211** may transmit a connection request (e.g., a SIP INVITE message) for the relay call to the service server **311** (e.g., the CSCF server/TAS). In an embodiment, the connection request for the relay call may include a device ID of the first electronic

device **211** as caller information, and a group ID of the communication group to which the first electronic device **211** belongs as callee information.

In operation **619**, the service server **311** may transmit a response corresponding to the connection request, and ringing to the first electronic device **211**. In an embodiment, the service server **311** may identify that at least one electronic device (e.g., second electronic device **1 221** and second electronic device **2 223**) is included in the communication group set for the first electronic device **211**. In an embodiment, the service server **311** may obtain information about the communication group set for the first electronic device **211** by querying the entitlement management server **411** in response to the connection request. In an embodiment, the service server **311** may pre-store the information about the communication group set for the first electronic device **211** according to the registration of the first electronic device **211** and read the information in response to the connection request.

In operation **621**, the service server **311** may transmit a connection request (e.g., a SIP INVITE message) for the relay call to second electronic device **1 221** in the communication group set in correspondence with the called phone number. In operation **623**, second electronic device **1 221** may transmit a response to the connection request and ringing to the service server **311**.

In operation **625**, the service server **311** may transmit a connection request (e.g., a SIP INVITE message) for the relay call to second electronic device **2 223** in the communication group. In operation **627**, second electronic device **2 223** may transmit a response to the connection request, and ringing.

Operations **629** to **647** may be a call termination and connection procedure.

Second electronic device **1 221** may terminate the call according to a user (callee) input in operation **629**, and transmit a response (e.g., a SIP OK message) indicating that the call has been terminated to the service server **311** in operation **631**. In operation **633**, the response may be transferred to the first electronic device **211**. In operation **635**, the first electronic device **211** may transmit an acknowledgment (ACK) indicating that the response has been received to the service server **311**. In operation **637**, the service server **311** may transfer the ACK to second electronic device **1 221**.

In operation **639**, the first electronic device **211** may transmit a response (e.g., a SIP OK message) to the peer electronic device **600** that has originated the call. In operation **641**, a call connection may be established between the peer electronic device **600** and the first electronic device **211**.

In operation **643**, a call session may be created between the first electronic device **211** and second electronic device **1 221** via the service server **311**, and a call connection may be established between the external (e.g., peer) electronic device **600** and second electronic device **1 221** through the call session. Media transmission may be performed among the first electronic device **211**, the service server **311**, and second electronic device **1 221** during the call connection.

When a call and message continuity service is used, a routing path may be established, for example, in the order of the external (e.g., peer) electronic device **600**→the first electronic device **211**→the service server **311** (e.g., the IMS server or the CSCF server/TAS)→the second electronic device or in the reverse order.

When receiving the response from second electronic device **1 221** in operation **631** and transmitting the ACK in

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operation 637, the service server 311 may transmit a cancellation request to second electronic device 223 to request to stop the connection request of operation 625 in operation 645 (e.g., see cancel at operation 645). Second electronic device 223 may stop ringing and transmit a response to the cancellation request to the service server 311 in operation 647 (e.g., see Response at operation 647 in FIG. 6).

FIG. 7 is a diagram illustrating a network environment according to an embodiment.

Referring to FIG. 7, the service server 311 may be configured to support the call and message continuity function between a plurality of first electronic devices (e.g., including the first electronic device 211 (a primary device (PD)) and a plurality of second electronic devices (e.g., including the second electronic device 221 (a secondary device (SD)). The first electronic device 211 may be associated with the second electronic device 221 based on the same user account. The second electronic device 221 may not include a SIM, and may originate or terminate a call or transmit or receive a message (e.g. an SMS or MMS message) through the service server 311, using the call function of the first electronic device 211.

The call and message continuity function enables the second electronic device 221, which is not capable of using a call or a message alone, to use the communication function (e.g., a call function or a message function) of the first electronic device 211 connected to the same user account through the service server 311. The call and message continuity function may be abused for malicious purposes such as spam calls, spam messages, or voice phishing. In many actual cases, a malicious user may place a plurality of second electronic devices (e.g., including the second electronic device 221) in the same space 711 (e.g., a room or a building), place a plurality of corresponding first electronic devices (e.g., including the first electronic device 211) in another space 713, and transmit a very large number of calls or messages to users (e.g., peer electronic devices 721 and 723) that do not want to receive, through the plurality of second electronic devices (e.g., including the second electronic device 221) using the call function of the first electronic devices (e.g., including the first electronic device 211).

For example, the plurality of second electronic devices (e.g., the second electronic device 221) may be located in another country and transmit multiple calls or messages to domestic peers (e.g., external electronic devices) through the plurality of first electronic devices (e.g., the first electronic device 211) located in Korea.

When the second electronic device 221 makes a call, using the call function of the first electronic device 211, the service server 311 according to an embodiment may determine whether a condition that multiple calls are originated to multiple call peers is satisfied by the second electronic device 221. When the condition is satisfied, the service server 311 may determine whether there are other external electronic devices around the second electronic device 221, which originate multiple calls to multiple call peers. When the above conditions are satisfied, the service server 311 may determine that the second electronic device 221 and the nearby external electronic devices are spam callers.

FIG. 8 is a diagram illustrating a procedure of detecting a spam call according to an embodiment.

Referring to FIG. 8, in an embodiment, the service server 311 may connect a call originated by the second electronic device 221 located in a space 811 (e.g., a room or a building) to the first electronic device 211 located in the same or different space 813. The first electronic device 211 may

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originate the call through a long-range wireless communication network (e.g., the second network 199). The second electronic device 221 may not include a SIM, and may originate the call using the call function of the first electronic device 211 through the service server 311. The service server 311 may determine that the second electronic device 221 is capable of originating the call using the call function of the first electronic device 211, based on user account information and user authentication information about the first electronic device 211 and the second electronic device 221 received from the user account server 413.

In an embodiment, the service server 311 may collect call log information about calls which are originated by the second electronic device 221 and connected to call peers (e.g., peer electronic devices 821 and 823) through the first electronic device 211. In an embodiment, the call log information may include at least one of a unique call number, a call type (incoming call, outgoing call, or missed call), a call time, account information (an account ID and/or an account name), secondary device information, primary device information, or call peer information (e.g. a phone number and/or device information). The user account information may include the user account shared by the first electronic device 211 and the second electronic device 221, and may be identified through the user account server 413.

The secondary device information may be included in the call log information, when a call is originated or terminated by the second electronic device 221 as a secondary device. According to an embodiment, the secondary device information may include device information (e.g., at least one of a model name, a model number, a serial number, or a MAC address) about the second electronic device 221. For a call connected, directly or indirectly, between the first electronic device 211 and a call peer without involving the secondary device, the call log information may not include the secondary device information. In an embodiment, when the secondary device information is not included or is set to an invalid value (e.g., '0' or empty) for a call included in the call log information, the service server 311 may determine that the secondary device is not involved in the call (e.g., has not originated or terminated the call). In an embodiment, when the secondary device information is included in the call log information for the call, the service server 311 may determine that the call has been originated or terminated by the secondary device.

In an embodiment, the service server 311 may count the number of calls originated by at least one secondary device (e.g., the second electronic device 221) related to the first electronic device 211 by analyzing call log information about the first electronic device 211, and determine whether the counted number of calls is greater than a specified threshold. When the number of calls is greater than the threshold, the service server 311 may detect external electronic devices located around the second electronic device 221.

In an embodiment, the service server 311 may collect and analyze call log information about the detected external electronic devices, and determine whether the number of calls originated by each of the external electronic devices is greater than the specified threshold. In an embodiment, the service server 311 may request and receive user account information about the detected external electronic devices from the user account server 413, and obtain call log information about the detected external electronic devices based on the user account information. Upon detection of at least one external electronic device having more outgoing calls than the threshold, the service server 311 may deter-

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mine the detected at least one external electronic device and the second electronic device 221 as spam callers.

FIG. 9 is a diagram illustrating an operation of searching for nearby devices according to an embodiment.

Referring to FIG. 9, the service server 311 may connect, directly or indirectly, a call originated by the second electronic device 221 located in a space 911 (e.g., a room or a building) to the first electronic device 211 located in the same or different space 913. The first electronic device 211 may originate the call through a long-range wireless communication network (e.g., the second network 199), and the second electronic device 221 may use the call function of the first electronic device 211 through the service server 311. The service server 311 may determine that the second electronic device 221 is capable of originating the call using the call function of the first electronic device 211, based on user account information and user authentication information about the first electronic device 211 and the second electronic device 221 received from the user account server 413.

In an embodiment, upon completion of an outgoing call from the second electronic device 221 using the call function of the first electronic device 211 (operation 901), the service server 311 may collect call log information including information about the outgoing call (operation 903). In an embodiment, the service server 311 may identify whether a first condition for determining whether the second electronic device 221 is a spam caller is satisfied by analyzing the call log information. In an embodiment, when the number of calls including outgoing calls from the first electronic device 211 or outgoing calls from the second electronic device 221 is greater than a specified threshold, the service server 311 may determine that the first condition is satisfied. In an embodiment, when the number of calls including outgoing calls from the second electronic device 221 except for outgoing calls from the first electronic device 211 is greater than the threshold, the service server 311 may determine that the first condition is satisfied. When the first condition is satisfied, the service server 311 may search for nearby devices (e.g., external electronic devices 921a, 921b, and 921c) located around the second electronic device 221.

In an embodiment, the service server 311 may search for and discover the external electronic devices 921a, 921b, and 921c using a nearby scan function of the second electronic device 221. In an embodiment, the service server 311 may transmit a scan request for requesting a nearby device search to the second electronic device 221 (operation 905), and receive nearby device information from the second electronic device 221 in response to the scan request (operation 907). The second electronic device 221 may discover the external electronic devices 921a, 921b, and 921c located in the vicinity, using a short-range communication technology (e.g., Bluetooth or WiFi) in response to the scan request, generate nearby device information about the external electronic devices 921a, 921b, and 921c, and transmit the nearby device information to the service server 311. The nearby device information may include, for example, MAC addresses (e.g., Bluetooth MAC addresses or WiFi MAC addresses) of the external electronic devices 921a, 921b, and 921c.

In an embodiment, the service server 311 may obtain user account information about the external electronic devices 921a, 921b, and 921c from the user account server 413, using the nearby device information as a key, and obtain call log information about the external electronic devices 921a, 921b, and 921c corresponding to the user account informa-

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tion from another server or read the call log information from internal memory (e.g., the memory 317).

According to an embodiment, the service server 311 may determine whether there is at least one external electronic device satisfying the afore-described first condition among the external electronic devices 921a, 921b, and 921c by analyzing the call log information about the external electronic devices 921a, 921b, and 921c. In an embodiment, when the number of calls originated by the external electronic device 921a is greater than the threshold, the service server 311 may determine that the external electronic device 921a satisfies the first condition. In an embodiment, when the external electronic device 921a is a secondary device, and the number of calls originated by the external electronic device 921a through a primary device (not shown) is greater than the threshold, the service server 311 may determine that the external electronic device 921a satisfies the first condition. In an embodiment, when the external electronic device 921a is a secondary device, and the total number of calls originated by the external electronic device 921a through the primary device (not shown) and calls originated by the primary device is greater than the threshold, the service server 311 may determine that the external electronic device 921a satisfies the first condition.

For example, the service server 311 may detect that the external electronic device 921a and the external electronic device 921b among the external electronic devices 921a, 921b, and 921c satisfy the first condition as a result of the determination. According to an embodiment, the service server 311 may determine that the second electronic device 221, the external electronic device 921a, and the external electronic device 921b are spam callers. In an embodiment, the service server 311 may store information indicating that the second electronic device 221, the external electronic device 921a, and the external electronic device 921b are spam callers in the internal memory (e.g., the memory 317). In an embodiment, the service server 311 may notify the user account server 413 and/or another server (e.g., the entitlement management server 411, the user authentication server, or a communication operator server) of the information indicating that the second electronic device 221, the external electronic device 921a, and the external electronic device 921b are spam callers. In response to the notification from the service server 311, a communication operator may perform a specified operation such as blocking the communication accounts of the second electronic device 221, the external electronic device 921a, the external electronic device 921b, and related primary devices (e.g., the first electronic device 211) or transmitting a warning message. In an embodiment, the service server 311 may notify a public institution such as the police of information about the spam callers.

In an embodiment, when the outgoing calls of the second electronic device 221 satisfy the first condition, the service server 311 may search for nearby devices around the first electronic device 211 and nearby devices around the second electronic device 221, and determine whether the first condition is satisfied for the nearby devices discovered as a result of the search. In an embodiment, the service server 311 may determine nearby devices satisfying the first condition as spam callers.

FIG. 10 is a diagram illustrating an operation of searching for nearby devices, using location information according to an embodiment.

Referring to FIG. 10, the service server 311 may connect a call originated by the second electronic device 221 located in a space 1011 (e.g., a room or a building) to the first

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electronic device **211** located in the same or different space **1013**. The first electronic device **211** may originate the call through a long-range wireless communication network (e.g., the second network **199**), and the second electronic device **221** may use the call function of the first electronic device **211** through the service server **311**. The service server **311** may determine that the second electronic device **221** is capable of originating the call using the call function of the first electronic device **211**, based on user account information and user authentication information about the first electronic device **211** and the second electronic device **221** received from the user account server **413**.

In an embodiment, upon completion of an outgoing call from the second electronic device **221** using the call function of the first electronic device **211**, the service server **311** may collect call log information including information about the outgoing call. In an embodiment, the service server **311** may identify that a condition (referred to as a “first condition”) that the number of recent outgoing calls is greater than a threshold is satisfied for the first electronic device **211** associated with the second electronic device **221** by analyzing the call log information. In an embodiment, the service server **311** may determine the first condition by comparing the threshold with the number of calls including outgoing calls from the first electronic device **211** or outgoing calls from the second electronic device **221**. In an embodiment, the service server **311** may determine the first condition by comparing the threshold with the number of calls including outgoing calls from the second electronic device **221** except for outgoing calls from the first electronic device **211**. When the first condition is satisfied, the service server **311** may search for nearby devices (e.g., external electronic devices **1021a**, **1021b**, and **1021c**) located around the second electronic device **221**.

In an embodiment, the service server **311** may search for the external electronic devices **1021a**, **1021b**, and **1021c** based on location information about the second electronic device **221**. In an embodiment, the location information may include a cell ID, access point information, and/or a GPS location (a latitude and a longitude). In an embodiment, when the second electronic device **221** is connected, directly or indirectly, to the service server **311** using WiFi, the location information may include access point information identifying an access point **1001** accessed by the second electronic device **221**.

In an embodiment, the service server **311** may discover electronic devices (e.g., the external electronic devices **1021a**, **1021b**, and **1021c**) connected, directly or indirectly, to the same access point **1001** as the second electronic device **221**. In an embodiment, the service server **311** may identify a list of a plurality of electronic devices (e.g., secondary devices) connected through the service server **311**, discover the external electronic devices **1021a**, **1021b**, and **1021c** having the same Internet protocol (IP) address as the second electronic device **221** and port numbers different from the second electronic device **221**, and obtain device information about the external electronic devices **1021a**, **1021b**, and **1021c**. The device information may include, for example, MAC addresses of the external electronic devices **1021a**, **1021b**, and **1021c**.

In an embodiment, the service server **311** may obtain user account information about the external electronic devices **1021a**, **1021b**, and **1021c** from the user account server **413**, using the device information as a key, and obtain call log information about the external electronic devices **1021a**, **1021b**, and **1021c** corresponding to the user account infor-

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mation from another server or read the call log information from the internal memory (e.g., the memory **317**).

According to an embodiment, the service server **311** may determine whether there is at least one external electronic device satisfying the afore-described first condition among the external electronic devices **1021a**, **1021b**, and **1021c** by analyzing the call log information about the external electronic devices **1021a**, **1021b**, and **1021c**. In an embodiment, when the number of calls originated by the external electronic device **1021a** is greater than the threshold, the service server **311** may determine that the external electronic device **1021a** satisfies the first condition. In an embodiment, when the external electronic device **1021a** is a secondary device, and the number of calls originated by the external electronic device **1021a** through a primary device (not shown) is greater than the threshold, the service server **311** may determine that the external electronic device **1021a** satisfies the first condition. In an embodiment, when the external electronic device **1021a** is a secondary device, and the total number of calls originated by the external electronic device **1021a** through the primary device (not shown) and calls originated by the primary device is greater than the threshold, the service server **311** may determine that the external electronic device **1021a** satisfies the first condition.

For example, the service server **311** may detect that the external electronic device **1021a** and the external electronic device **1021b** among the external electronic devices **1021a**, **1021b**, and **1021c** satisfy the first condition as a result of the determination. According to an embodiment, the service server **311** may determine that the second electronic device **221**, the external electronic device **1021a**, and the external electronic device **1021b** satisfying the first condition are spam callers. In an embodiment, the service server **311** may store information indicating that the second electronic device **221**, the external electronic device **1021a**, and the external electronic device **1021b** are spam callers in the internal memory (e.g., the memory **317**). In an embodiment, the service server **311** may notify the user account server **413** and/or another server (e.g., the entitlement management server **411**, the user authentication server, or a communication operator server) of the information indicating that the second electronic device **221**, the external electronic device **1021a**, and the external electronic device **1021b** are spam callers. In response to the notification from the service server **311**, the communication operator may perform a specified operation such as blocking the communication accounts of the second electronic device **221**, the external electronic device **1021a**, the external electronic device **1021b**, and related primary devices (e.g., the first electronic device **211**) or transmitting a warning message.

In an embodiment, when the outgoing calls of the second electronic device **221** satisfy the first condition, the service server **311** may search for nearby devices around the first electronic device **211** and nearby devices around the second electronic device **221**, and determine whether the first condition is satisfied for the nearby devices discovered as a result of the search. In an embodiment, the service server **311** may detect nearby devices located around the first electronic device **211**, using the cell ID or access point information about a base station accessed by the first electronic device **211**. In an embodiment, the service server **311** may determine nearby devices satisfying the first condition as spam callers.

FIG. **11** is a diagram illustrating a signal flow for an operation of detecting spam calls made by nearby devices according to an embodiment. In an embodiment, at least one

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of the following operations may be performed by the processor 313 of the service server 311.

Referring to FIG. 11, the second electronic device 221 (e.g., a secondary device) may be connected, directly or indirectly, to the first electronic device 211 (e.g., a primary device) through the service server 311, and make an outgoing call using the call function of the first electronic device 211. In operation 1100, the second electronic device 221 may notify the first electronic device 211 of completion of the call. According to an embodiment, the second electronic device 221 may notify the first electronic device 211 of the completion of the call by a SIP BYE message. In an embodiment, in operation 1100, the first electronic device 211 may detect that the second electronic device 221 has completed the call (e.g., pressing of a call end button or completion of the call by a call peer). In operation 1105, the first electronic device 211 may notify the service server 311 of the completion of the call. In an embodiment, the first electronic device 211 may generate a SIP BYE message and transmit the SIP BYE message to the service server 311, or may transfer the SIP BYE message received from the second electronic device 221 to the service server 311.

In operation 1110, the service server 311 may obtain and analyze call log information including calls performed by the second electronic device 221 to determine whether the second electronic device 221 is a spam caller. In an embodiment, the service server 311 may read call log information related to the second electronic device 221 from the memory 317 or request and obtain the call log information related to the second electronic device 221 from another server, when an outgoing call of the second electronic device 221 ends, every specified period, or when a specified condition is satisfied (e.g., the number or frequency of calls of the second electronic device 221 is greater than a specified threshold). In an embodiment, the service server 311 may read call log information related to the first electronic device 211 from the memory 317 or request and obtain the call log information related to the first electronic device 211 from another server, every specified period or when a specified condition (e.g., a network state or a request of a communication operator) is satisfied.

In operation 1115, the service server 311 may determine whether a first condition for determining a spam call ("SD spam call") of the secondary device is satisfied by analyzing the call log information. In an embodiment, when the second electronic device 221 is a secondary device, and the number of calls originated by the second electronic device 221 during a predetermined time period (e.g., several days or tens of days) is greater than a specified threshold, the service server 311 may determine that the first condition is satisfied. In an embodiment, when the second electronic device 221 is a secondary device, and the number of calls originated using the call function of the first electronic device 211 by the second electronic device 221 during a predetermined time period (e.g., several days or tens of days) is greater than the threshold, the service server 311 may determine that the first condition is satisfied. In an embodiment, when the second electronic device 221 is a secondary device, and the total number of calls originated using the call function of the first electronic device 211 by the second electronic device 221 and calls originated by the first electronic device 211 during a predetermined time period (e.g., several days or tens of days) is greater than the specified threshold, the service server 311 may determine that the first condition is satisfied.

In operation 1120, the service server 311 may search for nearby devices around the second electronic device 221 determined as satisfying the first condition. In an embodi-

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ment, the service server 311 may search for nearby devices (e.g., external electronic devices) located within a specified distance from the second electronic device 221 based on location information (e.g., a cell ID, access point information, or a GPS location) about the second electronic device 221. In an embodiment, the service server 311 may manage the locations of external electronic devices registered in the service server 311 or obtain location information about the external electronic devices from another server (e.g., the location server 417), and determine whether the external electronic devices are located around the second electronic device 221 by comparing the locations of the external electronic devices with the location information about the second electronic device 221.

In an embodiment, the service server 311 may determine that external electronic devices having the same cell ID or the same access point information as the second electronic device 221 are nearby devices of the second electronic device 221. In an embodiment, the service server 311 may determine that external electronic devices located within a specified distance from the location (e.g., GPS location) of the second electronic device 221 are nearby devices of the second electronic device 221.

In an embodiment, the service server 311 may obtain device information (e.g., MAC addresses) about the external electronic devices located around the second electronic device 221 through the search.

In operation 1125, the service server 311 may request account information (e.g., account IDs and/or account names) about one or more devices discovered as a result of the search in operation 1120 from the user account server 413, using device information (e.g., MAC addresses) about the external electronic devices. In operation 1130, the service server 311 may receive the account information from the user account server 413.

In operation 1135, the service server 311 may identify call log information about the one or more external electronic devices based on the account information. In an embodiment, the service server 311 may store and manage call log information about electronic devices registered in the service server 311, and identify the call log information about the one or more external electronic devices based on the account information.

In operation 1140, the service server 311 may detect at least one external electronic device satisfying the first condition among the one or more external electronic devices, and determine the second electronic device 221 and the detected at least one external electronic device as spam callers. In an embodiment, the service server 311 may detect at least one external electronic device having more outgoing calls than a specified threshold during a specified time among the one or more discovered external electronic devices. In an embodiment, the service server 311 may detect at least one external electronic device that is a secondary device and has more outgoing calls than the threshold during a specified time among the one or more discovered external electronic devices.

In operation 1145 that may be optional, the service server 311 may notify the user account server 413 and/or another server (e.g., the entitlement management server 411, the user authentication server, or the communication service provider server) of information about spam callers determined in operation 1140, for example, information about the second electronic device 221, the first electronic device 211, and the detected at least one external electronic device. In an embodiment, the service server 311 may block outgoing calls from the second electronic device 221, the first elec-

tronic device **211**, and the detected at least one external electronic device, which are determined to be spam callers.

FIG. **12** is a flowchart illustrating an operation of a service server for detecting spam calls made by nearby devices according to an embodiment. In an embodiment, at least one of the following operations may be performed by the processor **313** of the service server **311**.

Referring to FIG. **12**, the service server **311** may obtain and analyze first call log information in operation **1205**. According to an embodiment, the service server **311** may read first call log information related to the first electronic device **211** or the second electronic device **221** from the memory **317**, or request and obtain the first call log information related to the first electronic device **211** or the second electronic device **221** from another server, every specified time period or when a specified condition is satisfied (e.g., the number or frequency of calls made by the second electronic device **221** is greater than a specified threshold). In an embodiment, the service server **311** may obtain first call log information about the first electronic device **211** as a primary device related to the second electronic device **221** to determine whether the second electronic device **221** is a spam caller. The first call log information may include an outgoing call from the second electronic device **221**.

At least one of operation **1210** or operation **1215** may be an operation of determining whether the second electronic device **221** satisfies a first condition.

In operation **1210**, the service server **311** may determine whether the first call log information includes an outgoing call from the second electronic device **221** as a secondary device. In an embodiment, the service server **311** may recognize that the second electronic device **221** is a secondary device based on device information about the second electronic device **221**. In an embodiment, the service server **311** may identify the second electronic device **221** registered as a secondary device for the first electronic device **211**. In an embodiment, the service server **311** may determine that the second electronic device **221** is a secondary device based on secondary device information included in the first call log information. When the first call log information includes an outgoing call from the second electronic device **221** ('Yes' in operation **1210**), the procedure may proceed to operation **1215**. Otherwise ('No' in operation **1210**), the procedure may end. In an embodiment, operation **1210** may be skipped, and the service server **311** may proceed from operation **1205** to operation **1215**.

In operation **1215**, the service server **311** may count the number of outgoing calls from the second electronic device **221** (e.g., the number of outgoing calls during a predetermined time period) based on the first call log information, and determine whether the number of outgoing calls is greater than a specified threshold TH. In an embodiment, the service server **311** may compare the threshold with the total number of outgoing calls from the second electronic device **221** and outgoing calls directly from the first electronic device **211** as the primary device for the second electronic device **221**. In an embodiment, the service server **311** may compare the threshold with the total number of outgoing calls directly from the first electronic device **211** as the primary device for the second electronic device **221** and outgoing calls from at least one secondary device (e.g., the second electronic device **221**) related to the first electronic device **211**. When the number of outgoing calls is greater than the threshold ('Yes' in operation **1215**), the service server **311** may proceed to operation **1220**. Otherwise ('No' in operation **1215**), the procedure may end. Operations **1210**

and **1215** may correspond to the afore-described operation of determining whether the first condition is satisfied.

In operation **1220**, the service server **311** may search for nearby devices of the second electronic device **221**. In an embodiment, the service server **311** may search for nearby devices (e.g., external electronic devices) located within a specified distance from the second electronic device **221** based on location information (e.g., a cell ID, access point information, or a GPS location) about the second electronic device **221**. In an embodiment, the service server **311** may determine external electronic devices having the same cell ID or the same access point information as the second electronic device **221** as nearby devices of the second electronic device **221**. In an embodiment, the service server **311** may determine external electronic devices located within a specified distance from the location (e.g., GPS location) of the second electronic device **221** as nearby devices of the second electronic device **221**. In an embodiment, the service server **311** may discover one or more external electronic devices located within a specified distance from at least one of the first electronic device **211** or at least one secondary device (e.g., including the second electronic device **221**) related to the first electronic device.

In operation **1225**, the service server **311** may obtain account information (e.g., account IDs and/or account names) about one or more external electronic devices discovered as a result of the search in operation **1220** from the user account server **413**, using device information (e.g., MAC addresses) about the one or more external electronic devices. In operation **1230**, the service server **311** may obtain second call log information about the one or more external electronic devices based on the account information. In an embodiment, the service server **311** may store and manage call log information about electronic devices registered in the service server **311**, and identify the second call log information about the one or more external electronic devices based on the account information. According to an embodiment, the service server **311** may skip operation **1225**, and obtain the second call log information about the one or more external electronic devices, using the device information (e.g., MAC address) about the one or more discovered external electronic devices.

Operation **1235** may be an operation of determining whether each of the one or more external electronic devices satisfies the first condition.

In operation **1235**, the service server **311** may determine whether the number of outgoing calls from each of the one or more external electronic devices is greater than the specified threshold TH based on the second call log information about the one or more external electronic devices. In an embodiment, the service server **311** may compare the threshold with the total number of outgoing calls from each external electronic device and a corresponding primary device. Upon detection of at least one external electronic device having more outgoing calls than the threshold, the service server **311** may proceed to operation **1240**. Otherwise, the procedure may end. Operation **1235** may correspond to an operation of determining whether the afore-described first condition is satisfied for each discovered external electronic device.

In operation **1240**, the service server **311** may determine the second electronic device **221** and the at least one external electronic device detected in operation **1235** as spam callers. In an embodiment, the service server **311** may notify another server (e.g., the user account server **413**, the entitlement management server **411**, the user authentication server, or the communication operator server) of information about the

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second electronic device 221 and the detected at least one external electronic device. In an embodiment, the service server 311 may block calls made by the second electronic device 221 and the detected at least one external electronic device, and/or related primary devices.

In an embodiment, the service server 311 may determine whether the second electronic device 221 and at least one external electronic device located around it are spam message senders, using message log information (e.g., first message log information related to the second electronic device 221 and second message log information related the nearby device) similar to the above-described call log information.

FIG. 13 is a diagram illustrating a signal flow for an operation of detecting spam calls, using nearby scan according to an embodiment. In an embodiment, at least one of the following operations may be performed by the processor 313 of the service server 311.

Referring to FIG. 13, the second electronic device 221 (e.g., a secondary device) may be connected, directly or indirectly, to the first electronic device 211 (e.g., a primary device) through the service server 311, and make an outgoing call using the call function of the first electronic device 211. In operation 1300, the second electronic device 221 may notify the first electronic device 211 of completion of the call. In an embodiment, the second electronic device 221 may notify the first electronic device 211 of the completion of the call by a SIP BYE message. In an embodiment, in operation 1300, the first electronic device 211 may detect that the second electronic device 221 has completed the call (e.g., pressing of a call end button or completion of the call by a call peer). In operation 1305, the first electronic device 211 may notify the service server 311 of the completion of the call. In an embodiment, the first electronic device 211 may generate a SIP BYE message and transmit the SIP BYE message to the service server 311, or may transfer the SIP BYE message received from the second electronic device 221 to the service server 311.

In operation 1310, the service server 311 may obtain and analyze call log information including calls performed by the second electronic device 221 to determine whether the second electronic device 221 is a spam caller. In an embodiment, the service server 311 may read first call log information related to the second electronic device 221 from the memory 317 or request and obtain the first call log information related to the second electronic device 221 from another server, when an outgoing call of the second electronic device 221 ends, every specified period, or when a specified condition is satisfied (e.g., the number or frequency of calls of the second electronic device 221 is greater than a specified threshold). In an embodiment, the service server 311 may read first call log information related to the first electronic device 211 from the memory 317 or request and obtain the first call log information related to the first electronic device 211 from another server, every specified period or when a specified condition (e.g., a network state or a request of a communication operator) is satisfied.

In operation 1315, the service server 311 may determine whether a first condition for determining a spam call ("SD spam call") of the secondary device is satisfied by analyzing the first call log information. In an embodiment, when the second electronic device 221 is a secondary device, and the number of calls originated by the second electronic device 221 during a predetermined time period (e.g., several days or tens of days) is greater than a specified threshold, the service server 311 may determine that the first condition is satisfied. In an embodiment, when the second electronic

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device 221 is a secondary device, and the number of calls originated using the call function of the first electronic device 211 by the second electronic device 221 during a predetermined time period (e.g., several days or tens of days) is greater than the specified threshold, the service server 311 may determine that the first condition is satisfied. In an embodiment, when the second electronic device 221 is a secondary device, and the total number of calls originated using the call function of the first electronic device 211 by the second electronic device 221 and calls originated by the first electronic device 211 during a predetermined time period (e.g., several days or tens of days) is greater than the specified threshold, the service server 311 may determine that the first condition is satisfied.

In operation 1320, the service server 311 may transmit a scan request for searching for nearby devices around the second electronic device 221 determined as satisfying the first condition. In an embodiment, the scan request may be transferred from the service server 311 to the second electronic device 221 through the first electronic device 211 or without passing through the first electronic device 211. In operation 1320a, the second electronic device 221 may execute a nearby scan function in response to the scan request. In an embodiment, the second electronic device 221 may search for electronic devices located around the second electronic device 221 and discover one or more external electronic devices, by using a short-range communication technology (e.g., Bluetooth or WiFi). In operation 1325, the second electronic device 221 may transmit a scan response including nearby device information indicating the one or more external electronic devices discovered through the nearby scan. In an embodiment, the nearby device information may include a list (e.g., Near_MAC_list) of Bluetooth MAC addresses or WiFi MAC addresses of the one or more discovered external electronic devices.

In operation 1330, the service server 311 may request account information (e.g., account IDs and/or account names) about the one or more discovered devices from the user account server 413, using the nearby device information (e.g., MAC addresses) obtained from the scan response. In operation 1335, the service server 311 may receive the account information from the user account server 413.

In operation 1340, the service server 311 may identify second call log information about the one or more discovered external electronic devices based on the account information. In an embodiment, the service server 311 may store and manage call log information about electronic devices registered in the service server 311, and identify the second call log information about the one or more discovered external electronic devices based on the account information.

In operation 1345, the service server 311 may detect at least one external electronic device satisfying the first condition among the one or more discovered external electronic devices, and determine the second electronic device 221 and the at least one detected external electronic device as spam callers. In an embodiment, the service server 311 may detect at least one external electronic device having more outgoing calls than a specified threshold during a specified time among the one or more discovered external electronic devices. In an embodiment, the service server 311 may detect at least one external electronic device that is a secondary device and has more outgoing calls than the threshold during a specified time among the one or more discovered external electronic devices.

In operation 1350 that may be optional, the service server 311 may notify the user account server 413 and/or another

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server (e.g., the entitlement management server **411**, the user authentication server, or the communication service provider server) of information about the spam callers determined in operation **1345**, for example, information about the second electronic device **221**, the first electronic device **211**, and the at least one detected external electronic device. In an embodiment, the service server **311** may block outgoing calls from the second electronic device **221**, the first electronic device **211**, and the at least one detected external electronic device, which are determined as spam callers.

FIG. **14** is a flowchart illustrating an operation of a service server for detecting spam calls by nearby scan of an electronic device according to an embodiment. In an embodiment, at least one of the following operations may be performed by the processor **313** of the service server **311**.

Referring to FIG. **14**, the service server **311** may obtain and analyze first call log information in operation **1405**. In an embodiment, the service server **311** may read first call log information related to the first electronic device **211** or the second electronic device **221** from the memory **317**, or request and obtain the first call log information related to the first electronic device **211** or the second electronic device **221** from another server, every specified time period or when a specified condition is satisfied (e.g., the number or frequency of calls made by the second electronic device **221** is greater than a specified threshold). In an embodiment, the service server **311** may obtain first call log information about the first electronic device **211** as a primary device related to the second electronic device **221** to determine whether the second electronic device **221** is a spam caller. The first call log information may include an outgoing call from the second electronic device **221**.

At least one of operation **1410** or operation **1415** may be an operation of determining whether the second electronic device **221** satisfies a first condition.

In operation **1410**, the service server **311** may determine whether the first call log information includes an outgoing call from the second electronic device **221** as a secondary device. In an embodiment, the service server **311** may recognize that the second electronic device **221** is a secondary device based on device information about the second electronic device **221**. In an embodiment, the service server **311** may identify the second electronic device **221** registered as a secondary device for the first electronic device **211**. In an embodiment, the service server **311** may determine that the second electronic device **221** is a secondary device based on secondary device information included in the first call log information. When the first call log information includes an outgoing call from the second electronic device **221**, the procedure may proceed to operation **1415**. Otherwise, the procedure may end. In an embodiment, operation **1410** may be skipped, and the service server **311** may proceed from operation **1405** to operation **1415**.

In operation **1415**, the service server **311** may count the number of outgoing calls from the second electronic device **221** (e.g., the number of outgoing calls during a predetermined time period) based on the first call log information, and determine whether the number of outgoing calls is greater than a specified threshold TH. In an embodiment, the service server **311** may compare the threshold with the total number of outgoing calls from the second electronic device **221** and outgoing calls directly from the first electronic device **211** as the primary device for the second electronic device **221**. In an embodiment, the service server **311** may compare the threshold with the total number of outgoing calls directly from the first electronic device **211** as the primary device for the second electronic device **221** and

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outgoing calls from at least one secondary device (e.g., the second electronic device **221**) related to the first electronic device **211**. When the number of outgoing calls is greater than the threshold ('Yes' in operation **1415**), the service server **311** may proceed to operation **1420**. Otherwise ('No' in operation **1415**), the procedure may end. Operations **1410** and **1415** may correspond to the afore-described operation of determining whether the first condition is satisfied.

In operation **1420**, the service server **311** may transmit a scan request for requesting to search for nearby devices to the second electronic device **221**. In operation **1425**, the service server **311** may receive nearby device information indicating one or more external electronic devices discovered by nearby scan from the second electronic device **221**. In an embodiment, the nearby device information may include one or more MAC addresses. In an embodiment, the service server **311** may transmit a scan request to the first electronic device **211** and at least one secondary device (e.g., including the second electronic device **221**) related to the first electronic device **211**, and receive nearby device information indicating one or more external electronic devices in response to the scan request. In an embodiment, the nearby device information may include at least one external electronic device discovered by the first electronic device **211** and at least one external electronic device discovered by the second electronic device **221**.

In operation **1430**, the service server **311** may obtain account information (e.g., account IDs and/or account names) about the one or more discovered external electronic devices from the user account server **413**, using MAC addresses of the one or more external electronic devices included in the nearby device information. In operation **1435**, the service server **311** may obtain second call log information about the one or more external electronic devices based on the account information. In an embodiment, the service server **311** may store and manage call log information about electronic devices registered in the service server **311**, and identify the second call log information about the one or more external electronic devices based on the account information. In an embodiment, the service server **311** may skip operation **1430**, and obtain the second call log information about the one or more external electronic devices, using the device information (e.g., MAC address) about the one or more discovered external electronic devices.

Operation **1440** may be an operation of determining whether each of the one or more external electronic devices satisfies the first condition.

In operation **1440**, the service server **311** may determine whether the number of outgoing calls from each external electronic device is greater than a specified threshold TH based on the second call log information about the one or more external electronic devices. In an embodiment, the service server **311** may compare the threshold with the total number of outgoing calls from each external electronic device and a corresponding primary device. Upon detection of at least one external electronic device having more outgoing calls than the threshold, the service server **311** may proceed to operation **1445**. Otherwise, the procedure may end. Operation **1440** may correspond to an operation of determining whether the afore-described first condition is satisfied for each discovered external electronic device.

In operation **1445**, the service server **311** may determine the second electronic device **221** and the at least one external electronic device detected in operation **1440** as spam callers. In an embodiment, the service server **311** may notify another server (e.g., the user account server **413**, the entitlement

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management server **411**, the user authentication server, or the communication operator server) of information about the second electronic device **221** and the at least one detected external electronic device. In an embodiment, the service server **311** may block calls made by the second electronic device **221** and the at least one detected external electronic device, and/or related primary devices.

In certain example embodiments, the service server **311** may determine whether the second electronic device **221** and at least one external electronic device located around it are spam message senders, using message log information (e.g., first message log information related to the second electronic device **221** and second message log information about the nearby device) similar to the above-described call log information.

In certain example embodiments, the service server **311** may detect a spam caller who makes multiple calls, using call log information.

In certain example embodiments, the service server **311** may simultaneously detect a plurality of spam callers including nearby devices.

According to an embodiment, the service server **311** may include the communication circuit **315** and the at least one processor **313** (comprising processing circuitry). The at least one processor may be connected, directly or indirectly, to a first electronic device and a second electronic device, using the communication circuit. The at least one processor may be configured to obtain first call log information including at least one outgoing call made using a call function of the first electronic device by the second electronic device. The at least one processor may be configured to determine whether the second electronic device satisfies a first condition for determining that the second electronic device is a spam caller, based on the first call log information. The at least one processor may be configured to search for one or more external electronic devices located around the second electronic device, based on the first condition being satisfied. The at least one processor may be configured to obtain second call log information about the one or more external electronic devices. The at least one processor may be configured to determine at least one of the one or more external electronic devices or the second electronic device as a spam caller, based on the second call log information.

In an embodiment, if the second electronic device is a secondary device using the call function of the first electronic device through the server, and the number of outgoing calls made using the call function of the first electronic device by the second electronic device and/or the number of outgoing calls made by the first electronic device is greater than a given threshold, the at least one processor may be configured to determine that the first condition is satisfied for the second electronic device.

In an embodiment, the at least one processor may be configured to determine whether at least one of the one or more external electronic devices satisfies the first condition, based on the second call log information, and determine at least one external electronic device and the second electronic device that satisfy the first condition as spam callers.

In an embodiment, if a first external electronic device among the one or more external electronic devices is a secondary device using a call function of a corresponding primary device through the server, and the number of outgoing calls made using the call function of the corresponding primary device by the first external electronic device and/or the number of outgoing calls made by the corresponding primary device is greater than a given thresh-

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old, the at least one processor may be configured to determine that the first external electronic device satisfies the first condition.

In an embodiment, the at least one processor may be configured to transmit a scan request for requesting to search for a nearby device to the second electronic device, based on the first condition being satisfied, and receive nearby device information indicating the one or more external electronic devices from the second electronic device in response to the scan request.

In an embodiment, the nearby device information may include MAC addresses of the one or more external electronic devices.

In an embodiment, the at least one processor may be configured to discover the one or more external electronic devices based on location information about the second electronic device, and the location information may include at least one of a cell ID, an address of an access point accessed by the second electronic device, or a GPS location.

In an embodiment, the at least one processor may be configured to identify an access point accessed by the second electronic device, and discover the one or more external electronic devices accessing the server through the identified access point.

In an embodiment, the at least one processor may be configured to discover the one or more external electronic devices having the same IP address as the second electronic device and different port numbers from the second electronic device.

In an embodiment, the at least one processor may be configured to request account information about the one or more external electronic devices from a user account server, receive the account information about the one or more external electronic devices from the user account server, and obtain the second call log information based on the account information.

According to an embodiment, an operation method of the server **311** may include obtaining (**1205** or **1405**) first call log information including at least one outgoing call made using a call function of a first electronic device by a second electronic device connected, directly or indirectly, to the server. The method may include determining (**1210** and **1215** or **1410** and **1415**) whether the second electronic device satisfies a first condition for determining that the second electronic device is a spam caller, based on the first call log information. The method may include searching (**1220**, or **1420** and **1425**) for one or more external electronic devices located around the second electronic device, based on the first condition being satisfied. The method may include obtaining (**1230** or **1435**) second call log information about the one or more external electronic devices. The method may include determining (**1235** and **1240** or **1440** and **1445**) at least one of the one or more external electronic devices or the second electronic device as a spam caller, based on the second call log information.

In an embodiment, determining whether the second electronic device satisfies the first condition may include determining that the first condition is satisfied for the second electronic device, in response to the second electronic device being a secondary device using the call function of the first electronic device through the server, and the number of outgoing calls made using the call function of the first electronic device by the second electronic device and/or the number of outgoing calls made by the first electronic device being greater than a given threshold.

In an embodiment, determining the at least one of the one or more external electronic devices or the second electronic

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device as a spam caller may include determining whether at least one of the one or more external electronic devices satisfies the first condition, based on the second call log information, and determining at least one external electronic device and the second electronic device that satisfy the first condition as spam callers.

In an embodiment, determining whether the at least one of the one or more external electronic devices satisfies the first condition may include, in response to a first external electronic device among the one or more external electronic devices being a secondary device using a call function of a corresponding primary device through the server, and the number of outgoing calls made using the call function of the corresponding primary device by the first external electronic device and/or the number of outgoing calls made by the corresponding primary device being greater than a given threshold, determining that the first external electronic device satisfies the first condition.

In an embodiment, searching for the one or more external electronic devices located around the second electronic device may include transmitting a scan request for requesting to search for a nearby device to the second electronic device, based on the first condition being satisfied, and receiving nearby device information indicating the one or more external electronic devices from the second electronic device in response to the scan request. "Based on" as used herein covers based at least on.

In an embodiment, the nearby device information may include MAC addresses of the one or more external electronic devices.

In an embodiment, searching for the one or more external electronic devices located around the second electronic device may include discovering the one or more external electronic devices based on location information about the second electronic device, and the location information includes at least one of a cell ID, an address of an access point accessed by the second electronic device, or a GPS location.

In an embodiment, searching for the one or more external electronic devices located around the second electronic device may include identifying an access point accessed by the second electronic device, and discovering the one or more external electronic devices accessing the server through the identified access point.

Each embodiment herein may be used in combination with any other embodiment(s) described herein.

In an embodiment, searching for the one or more external electronic devices located around the second electronic device may include discovering the one or more external electronic devices having the same IP address as the second electronic device and different port numbers from the second electronic device.

In an embodiment, obtaining the second call log information about the one or more external electronic devices may include requesting account information about the one or more external electronic devices from a user account server, receiving the account information about the one or more external electronic devices from the user account server, and obtaining the second call log information based on the account information.

The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance.

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According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

It should be appreciated that various certain example embodiments and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as "A or B", "at least one of A and B", "at least one of A or B", "A, B, or C", "at least one of A, B, and C", and "at least one of A, B, or C", may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as "1st" and "2nd", or "first" and "second" may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term "operatively" or "communicatively", as "coupled with", "coupled to", "connected with", or "connected to" another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via at least a third element.

As used in connection with various certain example embodiments, the term "module" may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, logic, logic block, part, or circuitry. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC). Thus, each "module" herein may comprise circuitry.

Various embodiments as set forth herein may be implemented as software (e.g., the program 140) including one or more instructions that are stored in a storage medium (e.g., internal memory 136 or external memory 138) that is readable by a machine (e.g., the electronic device 101). For example, a processor (e.g., the processor 120) of the machine (e.g., the electronic device 101) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term 'non-transitory' simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

According to an embodiment, a method according to various certain example embodiments may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be

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distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

While the disclosure has been illustrated and described with reference to various embodiments, it will be understood that the various embodiments are intended to be illustrative, not limiting. It will further be understood by those skilled in the art that various changes in form and detail may be made without departing from the true spirit and full scope of the disclosure, including the appended claims and their equivalents. It will also be understood that any of the embodiment(s) described herein may be used in conjunction with any other embodiment(s) described herein.

The invention claimed is:

1. A server comprising:

memory storing instructions;
communication circuitry; and
at least one processor,

wherein the instructions, when executed by the at least one processor individually or collectively, cause the server to:

connect to a first electronic device and a second electronic device, through the communication circuitry,
obtain first call log information including at least one outgoing call made using a call function of the first electronic device by the second electronic device,

determine whether the second electronic device satisfies a first condition for determining that the second electronic device is a spam caller, based on the first call log information,

based on the first condition being satisfied, transmit, to the second electronic device, a scan request for requesting a nearby device scanning, through the communication circuitry,

receive, from the second electronic device, nearby device information about one or more external electronic devices located around the second electronic device, through the communication circuitry, after transmitting the scan request, wherein the nearby device information includes one or more media access control (MAC) addresses of the one or more external electronic devices,

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obtain second call log information about the one or more external electronic devices, and

determine at least one of the one or more external electronic devices and/or the second electronic device as a spam caller, based on the second call log information.

2. The server of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the server to determine that the first condition is satisfied for the second electronic device based on the second electronic device being a secondary device using the call function of the first electronic device through the server, and a number of outgoing calls made using the call function of the first electronic device by the second electronic device and/or a number of outgoing calls made by the first electronic device being greater than a given threshold.

3. The server of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the server to:

determine whether at least one of the one or more external electronic devices satisfies the first condition, based on the second call log information, and

determine at least one external electronic device and the second electronic device that satisfy the first condition as spam callers.

4. The server of claim 3, wherein the instructions, when executed by the at least one processor individually or collectively, cause the server to determine that a first external electronic device satisfies the first condition based on the first external electronic device among the one or more external electronic devices being a secondary device using a call function of a corresponding primary device through the server, and a number of outgoing calls made using the call function of the corresponding primary device by the first external electronic device and/or a number of outgoing calls made by the corresponding primary device being greater than a given threshold.

5. The server of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the server to discover the one or more external electronic devices based on location information about the second electronic device,

wherein the location information includes at least one of:
a cell identifier (ID), an address of an access point accessed by the second electronic device, or a global positioning system (GPS) location.

6. The server of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the server to:

identify an access point accessed by the second electronic device, and

discover the one or more external electronic devices accessing the server through the identified access point.

7. The server of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the server to discover the one or more external electronic devices having the same Internet protocol (IP) address as the second electronic device and different port numbers from the second electronic device.

8. The server of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the server to:

request account information about the one or more external electronic devices from a user account server,

receive the account information about the one or more external electronic devices from the user account server, and

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obtain the second call log information based on the account information.

9. An operation method of a server, the method comprising:

obtaining first call log information including at least one outgoing call made using a call function of a first electronic device by a second electronic device connected to the server;

determining whether the second electronic device satisfies a first condition for determining that the second electronic device is a spam caller, based on the first call log information;

based on the first condition being satisfied, transmitting, to the second electronic device, a scan request for requesting a nearby device scanning;

receiving, from the second electronic device, nearby device information about one or more external electronic devices located around the second electronic device, after transmitting the scan request, wherein the nearby device information includes one or more media access control (MAC) addresses of the one or more external electronic devices;

obtaining second call log information about the one or more external electronic devices; and

determining at least one of the one or more external electronic devices or the second electronic device as a spam caller, based on the second call log information.

10. The operation method of claim 9, wherein determining whether the second electronic device satisfies the first condition comprises:

determining that the first condition is satisfied for the second electronic device, based on the second electronic device being a secondary device using the call function of the first electronic device through the server, and a number of outgoing calls made using the call function of the first electronic device by the second electronic device and/or a number of outgoing calls made by the first electronic device being greater than a given threshold.

11. The operation method of claim 9, wherein determining the at least one of the one or more external electronic devices or the second electronic device as a spam caller comprises:

determining whether at least one of the one or more external electronic devices satisfies the first condition, based on the second call log information, and

determining at least one external electronic device and/or the second electronic device that satisfy the first condition as spam callers.

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12. The operation method of claim 11, wherein determining whether the at least one of the one or more external electronic devices satisfies the first condition comprises:

determining that a first external electronic device satisfies the first condition, based on the first external electronic device among the one or more external electronic devices being a secondary device using a call function of a corresponding primary device through the server, and a number of outgoing calls made using the call function of the corresponding primary device by the first external electronic device and/or a number of outgoing calls made by the corresponding primary device being greater than a given threshold.

13. The operation method of claim 9, wherein searching for the one or more external electronic devices located around the second electronic device comprises discovering the one or more external electronic devices based on location information about the second electronic device, and

wherein the location information includes at least one of a cell identifier (ID), an address of an access point accessed by the second electronic device, or a global positioning system (GPS) location.

14. The operation method of claim 9, wherein searching for the one or more external electronic devices located around the second electronic device comprises:

identifying an access point accessed by the second electronic device, and

discovering the one or more external electronic devices accessing the server through the identified access point.

15. The operation method of claim 9, wherein searching for the one or more external electronic devices located around the second electronic device comprises discovering the one or more external electronic devices having the same Internet protocol (IP) address as the second electronic device and different port numbers from the second electronic device.

16. The operation method of claim 9, wherein obtaining the second call log information about the one or more external electronic devices comprises:

requesting account information about the one or more external electronic devices from a user account server, receiving the account information about the one or more external electronic devices from the user account server, and

obtaining the second call log information based on the account information.

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